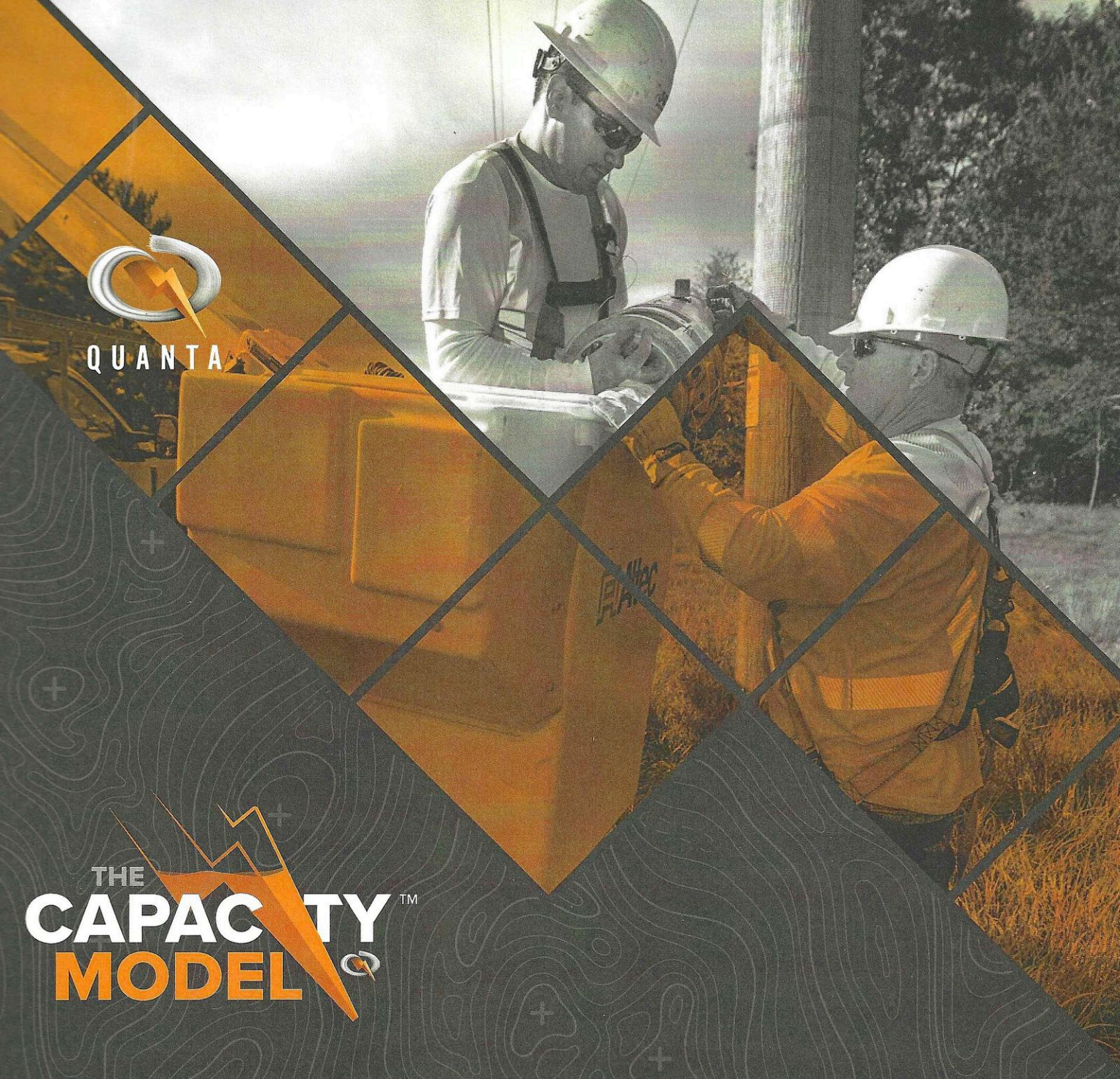

QUANTA

THE CAPACITY MODEL™

STKY AND THE ENERGY WHEEL FOR THE POWER AND TELECOMMUNICATIONS INDUSTRY

INSTRUCTOR MANUAL

V 1.0

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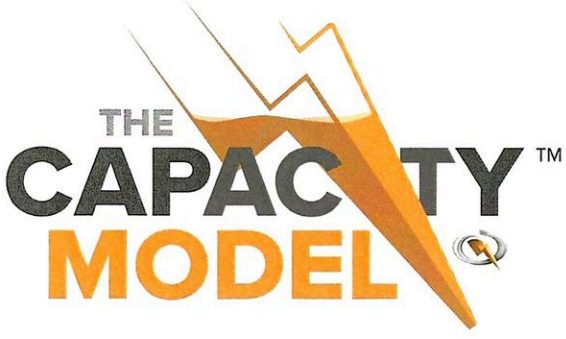
THE
CAPACITY
MODEL
TM

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STKY and the Energy Wheel
for the Power and Telecommunications Industry

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TCM 3004

STKY AND THE ENERGY WHEEL FOR THE POWER AND TELECOMMUNICATIONS INDUSTRY

SYLLABUS

COURSE DESCRIPTION

Trainees learn how to identify events that result from the release of energy and to identify and categorize workplace hazards.

The class is taken in conjunction with the Quanta Human Performance Principles module. Complete both modules within three months to gain maximum learning and retention benefits.

Intended audience: Operations management, safety professionals, supervisors who direct the work of others, and those responsible for leading job briefings.

LEARNING OUTCOMES

1.a. Define a STKY control.
1.b. Define a STKY event, STKY potential, STKY actual, and a STKY success.
1.c. Apply the STKY Event Classification Tool to practical scenarios.
2.a. Use the Energy Wheel to identify hazards associated with a task.
2.b. Identify preventive and mitigative controls to prevent and reduce the consequences of unwanted events.

SUGGESTED PREREQUISITES

- Understanding The Capacity Model®

COURSE MATERIALS

- Materials to be issued at the beginning of the course.

COURSE OUTLINE

- Housekeeping
- STKY
- The Energy Wheel
- Exam

INSTRUCTIONAL METHODS

- Lecture
- Learning activities
- Knowledge check

METHODS OF MEASUREMENT

KNOWLEDGE
Written exam:
A = 100%–90%
B = 89%–80%
C = 79%–70%
D = 69%–60%
F = 59% and below

If a student earns less than a 70% on the exam, one retake is permitted. If they do not earn at least a 70% on the exam retake, they must retake the course.

SKILL
Completion of all activities as directed by the instructor.

BEHAVIOR
Active participation and engagement in class discussions and activities.

STUDENT-TO-INSTRUCTOR RATIO
Maximum of 24:1.

CLOCK HOURS
5 Hours

POLICIES AND EXPECTATIONS

Building a positive learning environment—Be sure to share, support your classmates, and contribute to the positive learning environment during your course. You bring incredible value to the group and to the course. Please refrain from any behavior that you know doesn't work in a classroom environment.

TCM 3004

STKY AND THE ENERGY WHEEL FOR THE POWER AND TELECOMMUNICATIONS INDUSTRY

AGENDA

SCHEDULE

HOUSEKEEPING	*15 minutes*
STKY	*1 hour, 30 minutes*
ENERGY WHEEL	*2 hours, 45 minutes*
EXAM	*30 minutes*

Breaks are scheduled at the discretion of the instructor.

The lunch schedule is determined by the class start time and the option to complete the Quanta Human Performance Principles module in the same day.

USER GUIDE

INTRODUCTION

The slides and notes presented in this manual may look slightly different than those in the PowerPoint. They include additional instructions and support to help you teach the course. The slides images include symbols that align with symbols in the corresponding notes to help you quickly see what needs to be done during class. See Table 1.1 for a list of the symbols and their meaning.

Read through this manual before teaching the course, and have the notes with you during the lesson. Use this manual to record your training notes, application activities, stories, and so forth alongside the appropriate notes in this manual.

SYMBOLS

SYMBOL	MEANING
	Stop here during the presentation to ask the corresponding discussion question.
	Tell a story about the topic indicated in the instructor notes.
	Perform a formative assessment to guide your instruction.
	Complete one or more of the provided application activities.
	Look here for a helpful teaching tip.
	This is a good place to review the guiding question. There is no need for discussion. Simply remind the students of the guiding question.

Table 1.1 The e symbols are used on the slide images and corresponding notes to provide additional guidance for teaching the course.

MODULE OVERVIEW

GUIDING QUESTIONS

The guiding question is a focus of the lesson. Present it to the trainees at the beginning of the course and reinforce it throughout the training. Repetition is key in learning and this simple question reinforces the importance of the topic.

- Why do we need to be hyper-focused on STKY?
- Why do we need to be able to identify all of the Energy Wheel hazards?

KEY TERMS

capacity	The actual or potential ability to perform, yield, or withstand. With The Capacity Model, the ability to perform work safely, yield safe results, and withstand the outcome of an accident or mistake.
capacity for failure	In a work setting, the ability to perform a task while safely withstanding the outcome of an accident or mistake.
control	Acts (things a person does), objects (a device that works without acts), or systems (a combination of acts and objects) intended to prevent or mitigate an unwanted event.
job briefing	A required meeting held at a minimum before every job or shift to ensure every crewmember understands specific factors and responsibilities for that job. Also referred to as a tailboard.
mitigative control	Act, object, or system designed to reduce the consequences of an unwanted event. Mitigative controls help after the bad stuff has already happened.
preventive control	Act, object, or system designed to prevent an unwanted event. Preventive controls stop the bad stuff from happening
Quanta Energy Wheel	A hazard-identification tool that helps workers systematically look for types of energy. The Energy Wheel focuses on hazards that can cause a life-threatening, life-altering, or life-ending injury.
STKY	Stands for stuff that kills you. A source of energy that, when released and not controlled, can be a life-altering, life-threatening, or life-ending injury.
STKYa	*A* stands for "actual." A work incident (STKY event) that resulted in an injury that was life-threatening, life-altering, or life-ending.

MODULE OVERVIEW (cont.)

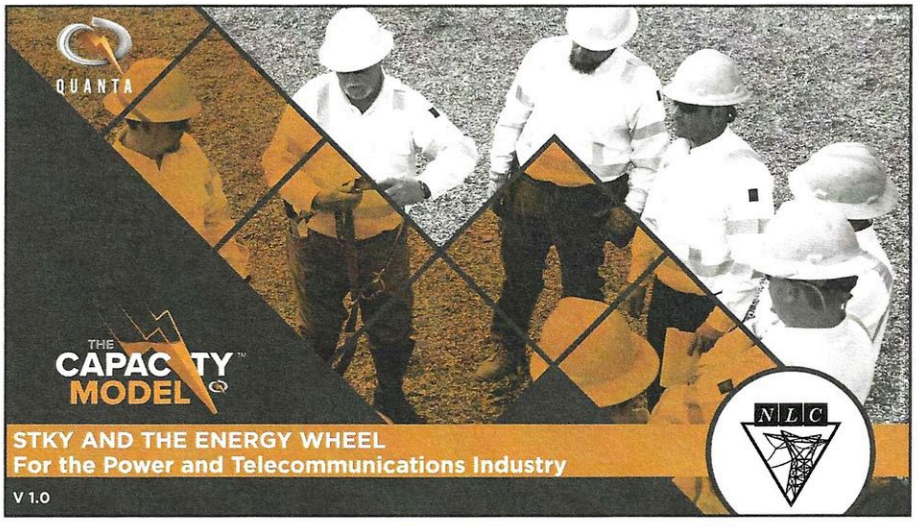
KEY TERMS (cont.)

STKY control	A control that prevents or mitigates a life-threatening, life-altering, or life-ending event (STKY event). Like a standard control, it may be an act, object, or system.
STKY discussion	Conversations meant to generate an understanding of how to identify STKY using the Energy Wheel. STKY discussions happen before the start of a job and as conditions change that may impact worker safety.
STKY event	An event where preventive controls failed, resulting in the unexpected or uncontrolled release of energy that could have reasonably resulted in a life-threatening, life-altering, or life-ending event.
STKYp	*P* stands for "potential." A STKY event in which no one was seriously injured but controls were either not in place, not functioning, or not effective. The only thing protecting the crew from a STKYa was luck.
STKY success	A situation where controls were in place, functioning, and effective. These controls successfully prevented a STKY event from becoming a STKYa or STKYp.
unwanted event	An event with a potential for harm from exposure to a hazard that was not adequately controlled—it is the bad stuff that can happen because of a hazard.

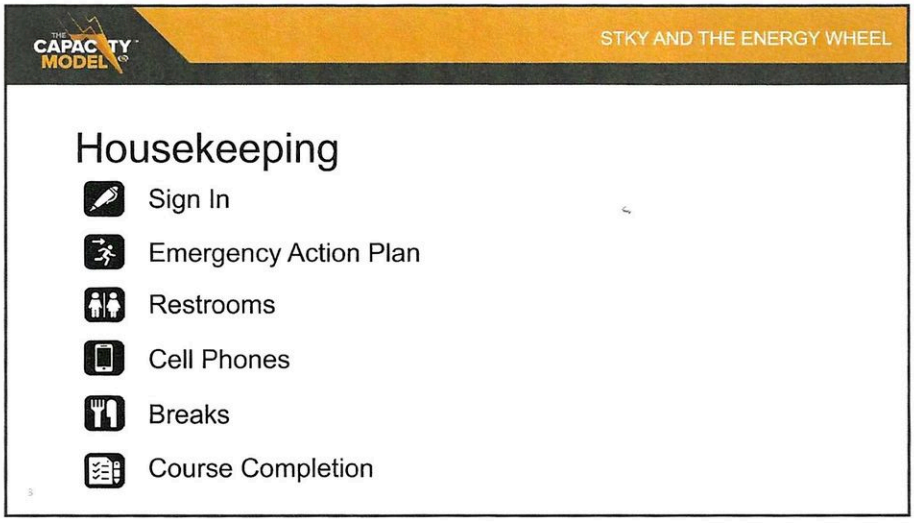
POWERPOINT PRESENTATION GUIDE

Use the corresponding notes below each slide image for additional guidance on delivering the content.

Show this slide while trainees are arriving.

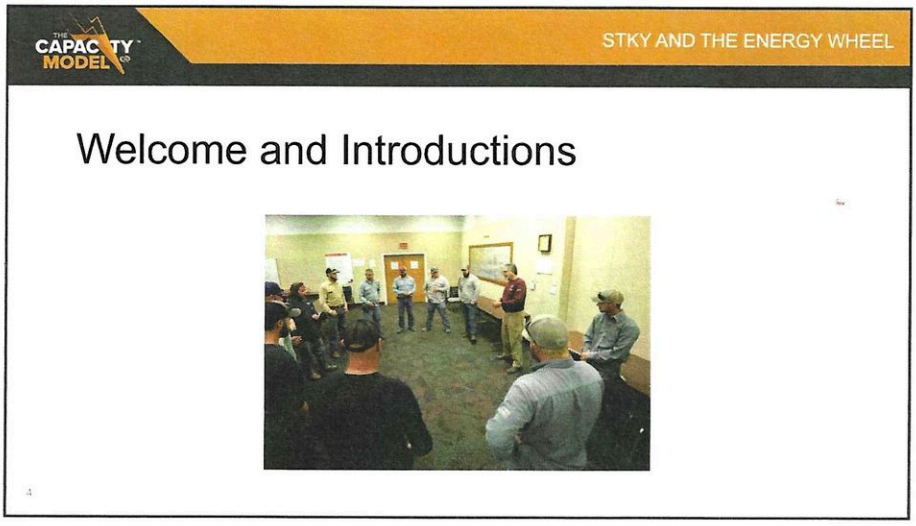


Briefly introduce yourself and welcome the trainees to the training.



1. Pass around a sign-in sheet and ensure that all trainees sign in.
2. Complete the emergency action plan:
   - Point out the emergency exits.
   - Explain where to meet up in case of an emergency.
   - Ask trainees who are trained in first aid and CPR to identify themselves. Choose one to be responsible for first aid and CPR and an alternate, if possible, in case the primary person is involved in the emergency.
   - Explain the location of the nearest AED.
   - Provide your cell number in case of an emergency.
   - Post the address of the training location in the room.
   - Identify one person to meet EMS in an emergency and an alternate in case the primary person assigned is involved in the emergency.
3. Provide the location of the nearest restrooms.
4. Explain the cell phone policy: Put on silent, take necessary calls outside the classroom.
5. Explain breaks and lunch (if a lunch break is applicable)..
   - You can give breaks anytime you think trainees need one.
   - A good rule of thumb is to provide a 5-10 minute break at least every hour or two, depending on the level of engagement and the pace.
   - Look for signs of disengagement, slouching posture, etc. and provide breaks accordingly.
6. Explain Human Performance Principles is one module of the course. The other module is STKY and the Energy Wheel.
   - To complete this module, trainees must participate in the class and earn at least a 70% on the exam. Explain the exam is 10 multiple-choice questions.

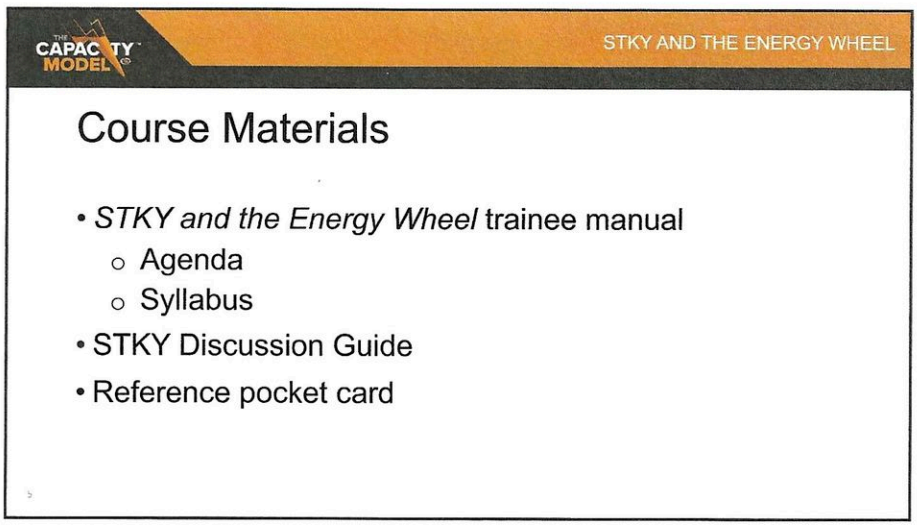
Identify a student who will advise you when the class may need a break.



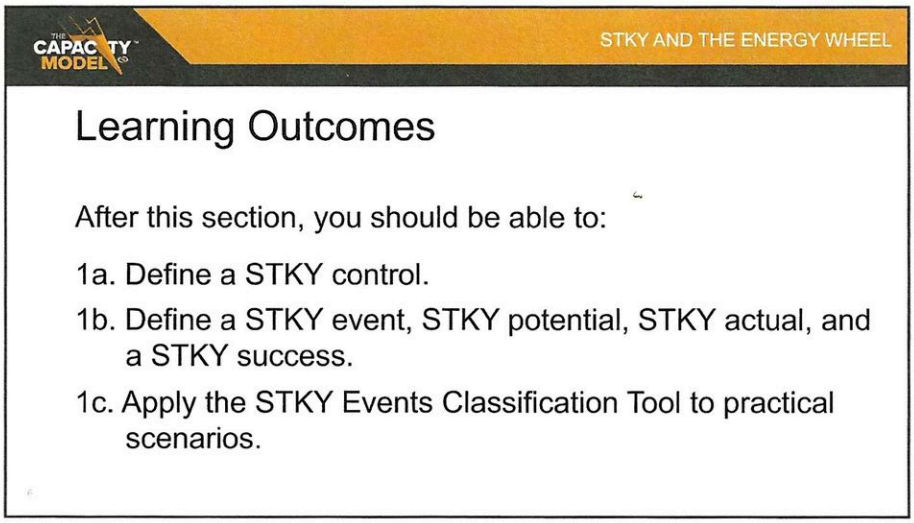
- Introduce yourself and share your background.
- Ask trainees to introduce themselves and share their backgrounds.
- Use this beginning segment to build rapport and establish credibility (positive learning environment).

Consider these questions to help with introductions.

- Who had the greatest influence in your career?
- Where was the most interesting place you have worked?
- What quality makes you good at your job?
- What does it mean to be safe?

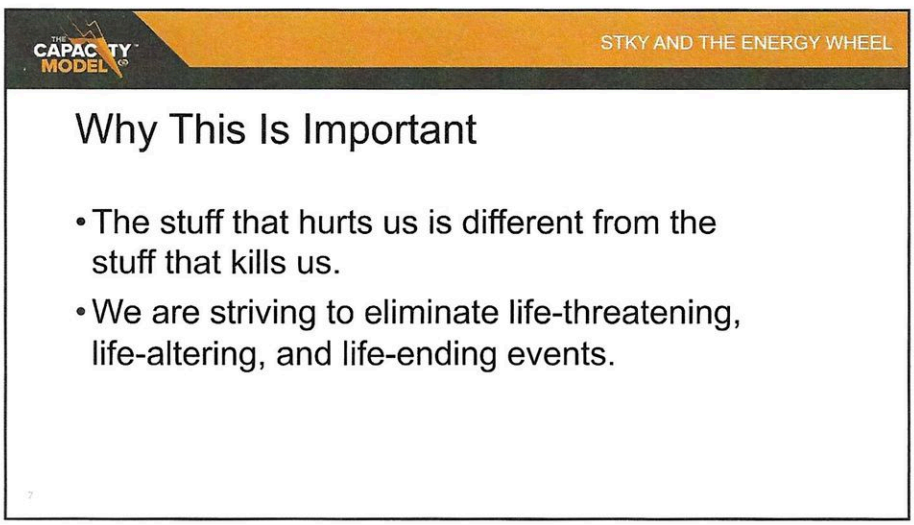


- Confirm trainees received a manual.
- Point out the syllabus and agenda on pages 4 and 5 of the manual.
- Confirm trainees received the STKY Discussion Guide and the TCM pocket card.



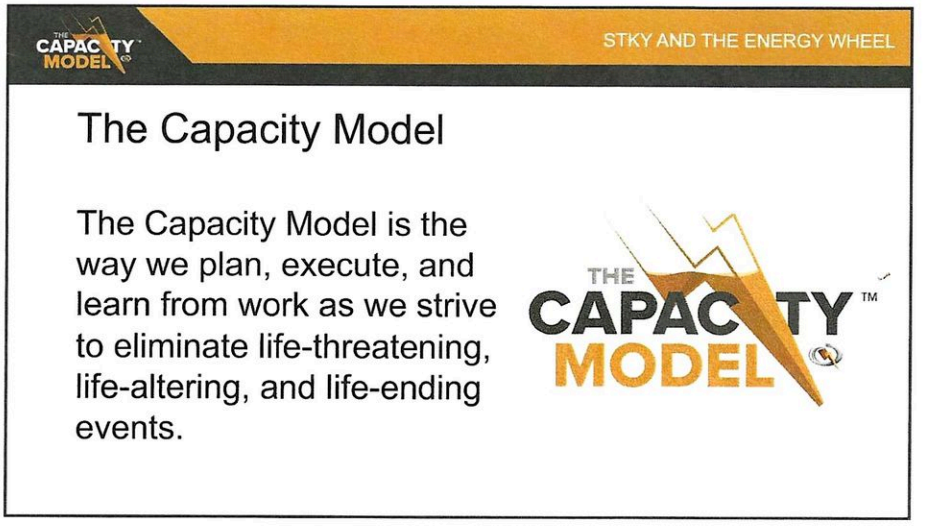
Slide aligns with page 7 of the trainee manual.

- Review learning outcomes.
- Read or ask for volunteers to read the learning outcomes.
- Write the learning outcomes on a whiteboard or post them in a visible area of the room to refer to during the training.



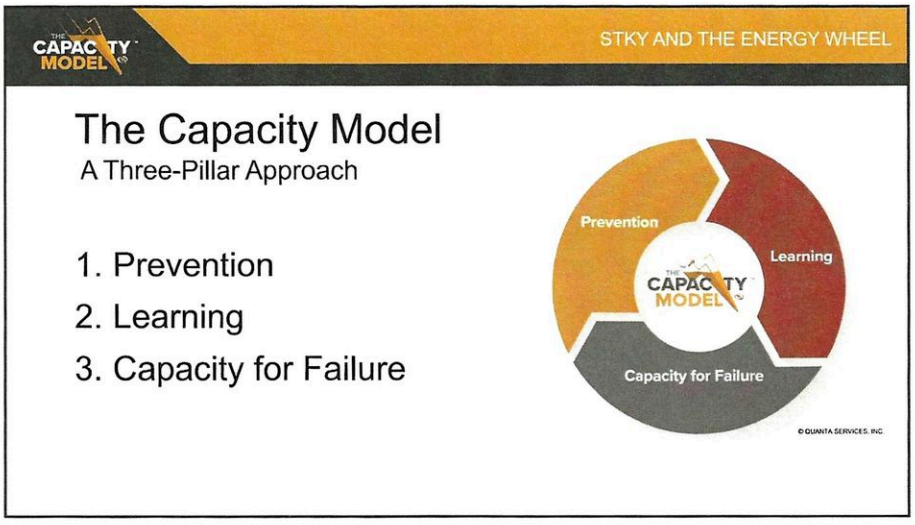
Use the bullet items to discuss why this course is important for the trainees.

Introduce the guiding questions:
Why do we need to be hyper-focused on STKY?
Why do we need to be able to identify all of the Energy Wheel hazards?



This slide is a duplicate from the Human Performance Principles module. If teaching both modules in the same day, the slide can be reviewed or skipped.

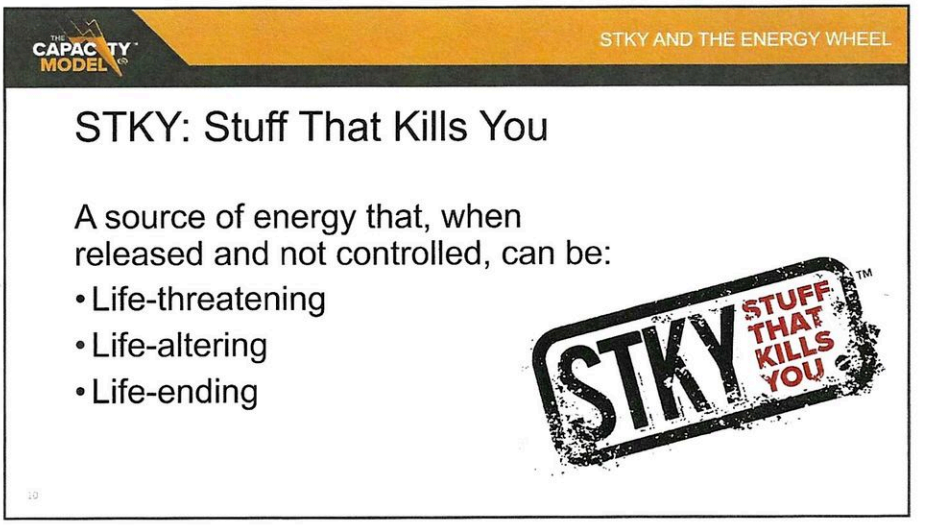
- Read the definition on the slide.
- Break down each element of the definition. For example:
- Eliminate life-threatening, life-altering, and life-ending events: While we always want to eliminate and mitigate all events that can cause injuries, The Capacity Model focuses on STKY events – the stuff that kills you.
- Plan work: Everything we do before the work begins allows us to focus on the tasks we perform and the preventive controls that are necessary.
- Execute work: But remember, prevention is not enough. As people execute the work in real time, they are continuously adapting and adjusting to normal conditions and variability of the job. We know that sometimes, things do go wrong, so we will always plan our work as if failure will happen today and build in the capacity to fail safely when it does.
- Learn from work: Whether learning from when things go wrong, or from things that go right, we know that learning is a deliberate improvement strategy. The only way to get better is to take the time to learn from those who do the work.



This slide is a duplicate from the Human Performance Principles module. If teaching both modules in the same day, the slide can be reviewed or skipped.

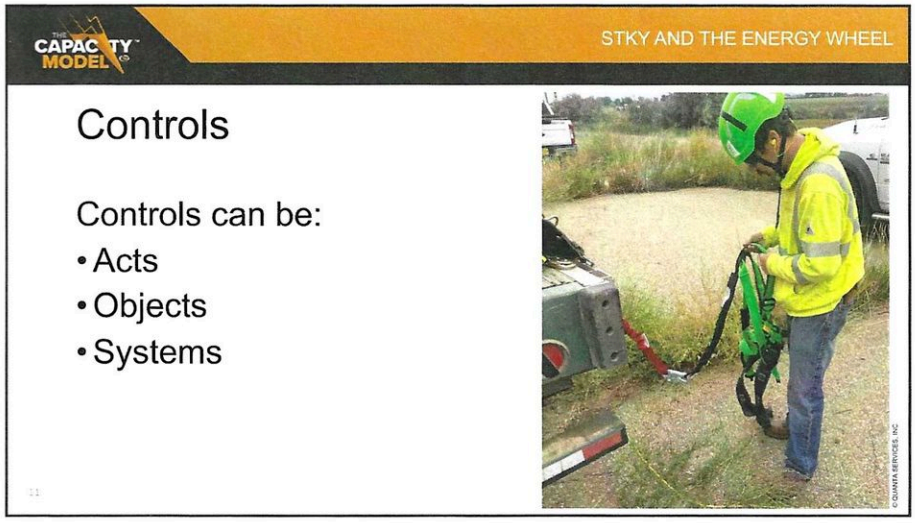
PRESENT THE FOLLOWING:

- The Capacity Model is a three-pillar approach to safety: prevention, learning, and capacity for failure.
- Prevention: Prevention is the traditional safety defense, and it has worked well. But there is still work we need to do to ensure everyone gets home safely every day; after all, People Make Mistakes (Principle 1).
- Learning: By creating an environment where we seek to learn from errors and incidents, instead of punishing our people, we can create trusting workplaces and uncover the true conditions that surround errors. We must view learning as a Deliberate Improvement Strategy (Principle 7).
- Capacity for Failure: Knowing that errors and failures are always around us, we must build resilience toward error with capacity. We must build capacity to safely absorb errors by implementing controls so that no one is injured when energy is released.



Slide aligns with page 8 of the trainee manual.

- Review the definition of STKY and the definitions of:
  - Life-threatening: Requires medical intervention to survive.
  - Life-altering: Results in loss of a body part or function.
  - Life-ending: Results in death.
- Explain you will cover STKY and STKY events in upcoming slides.
- Explain you will review types of controls first.



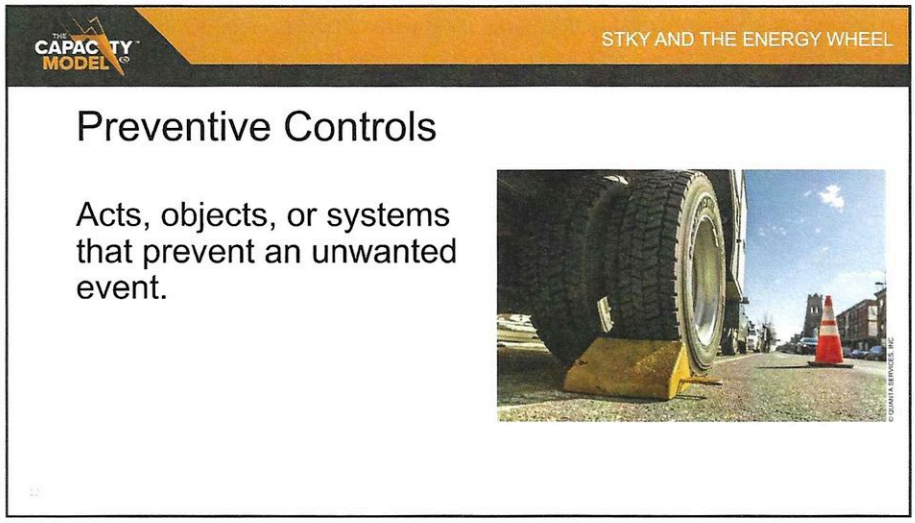
Slide aligns with page 9 of the trainee manual.

PRESENT THE FOLLOWING:

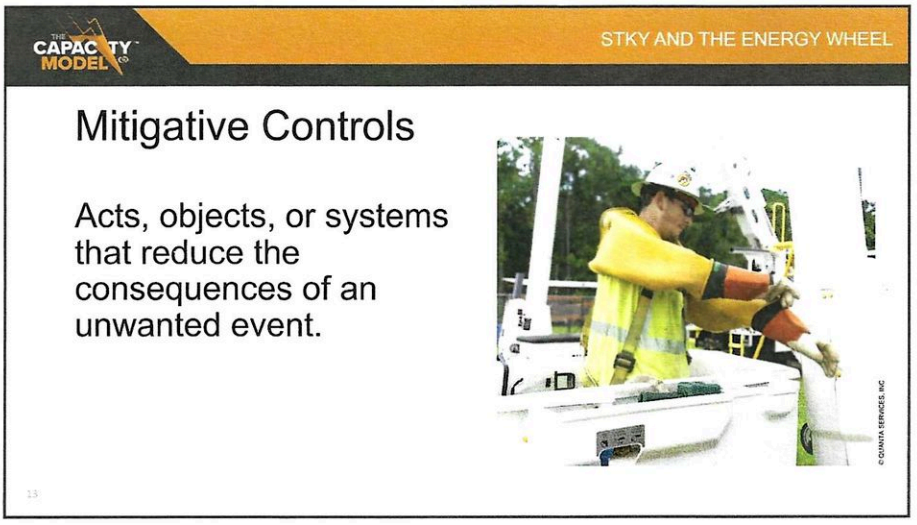
- Think about various job sites where you work and consider the "stuff that can kill you" in those surroundings.
- How do you put safety measures in place?
- By planning ahead and implementing controls, you can focus on ways to prevent unwanted STKY events and create an environment where you can reduce the consequences of those events.

Provide definitions of each:

- Acts: Things a person does (inspecting your harness or PPE)
- Objects: A device that works without acts (harness or PPE)
- Systems: A combination of acts and objects



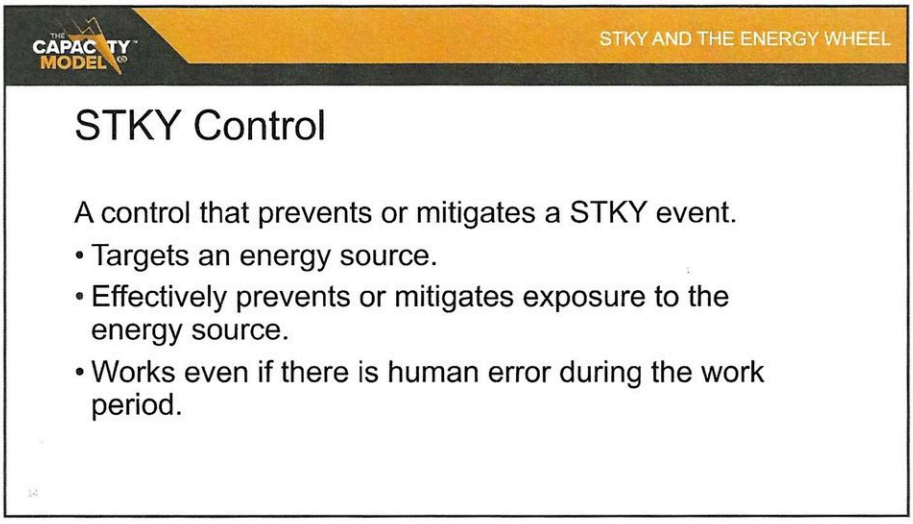
- Review the definition of preventive controls.
- Add to the definition: Preventive controls stop the bad stuff from happening.
- Provide examples of preventive controls that are acts, objects, and systems:
  - Acts: Maintaining minimum approach distance to prevent electrocution.
  - Objects: Using wheel chocks to prevent uncontrolled movement of a vehicle, trailer, etc.
  - Systems: Creating an equipotential zone to prevent electrocution.
- To determine if a control is preventive, ask the question: How does the control prevent the unwanted event?



- Review the definition of mitigative controls.
- Add to the definition: Mitigative controls help after the bad stuff has already happened.
- Reinforce that mitigative controls build capacity to reduce the consequences of an unwanted event when preventive controls fail.
- Recall the phrase: It's not "if" but "when" a failure occurs.

- Provide examples of mitigative controls that are acts, objects, and systems:
  - Acts: Posting a qualified observer at the entrance of a confined space.
  - Objects: Wearing PPE like hard hats, safety glasses, insulated gloves, and oxygen monitors.
  - Systems: Attaching tools to lanyards when working at heights
- To determine if a control is mitigative, ask the question: How does the control minimize the consequences after an unwanted event occurs?

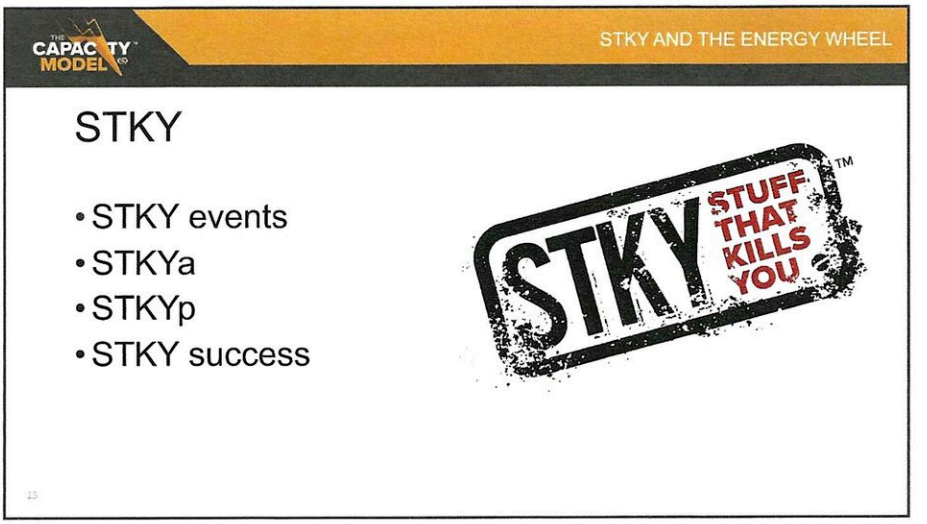
Explain BOTH preventive and mitigative controls are needed to ensure a safe work environment and to ensure everyone goes home safely at the end of the day.



Slide aligns with page 10 of the trainee manual.

- Review the definition of a STKY control.
- Direct trainees to page 10 to review the trenching example for STKY controls.
- Walk through the photos on page 10 related to the STKY controls example.
  - Installing shielding/shoring.
  - Placing a ladder/emergency egress.
  - Keeping spoils at least two feet from the trench edge.

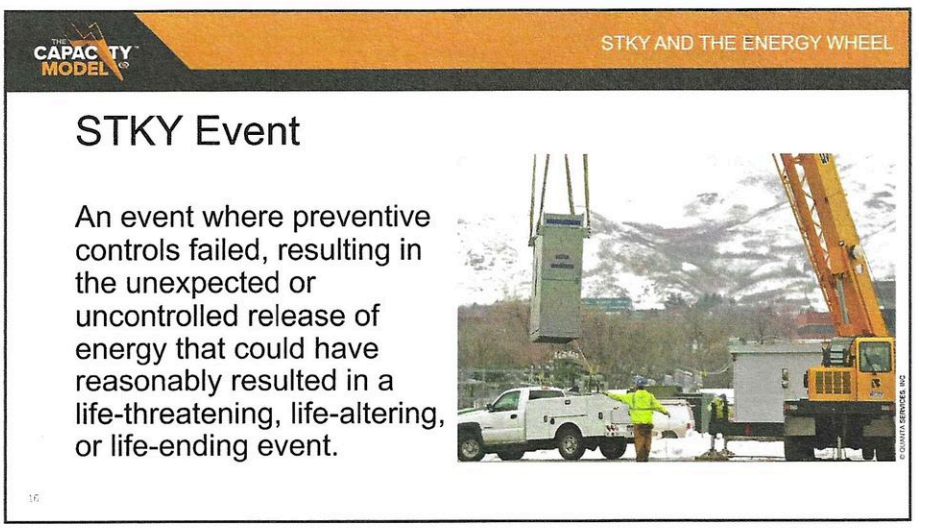
- Note that you may get varying answers on mitigative vs. preventive controls when reviewing the photos in the manual. It is healthy to have a debate about the types of controls. Direct the focus back to building in layers of controls and reinforce that preventive and mitigative controls must be present.



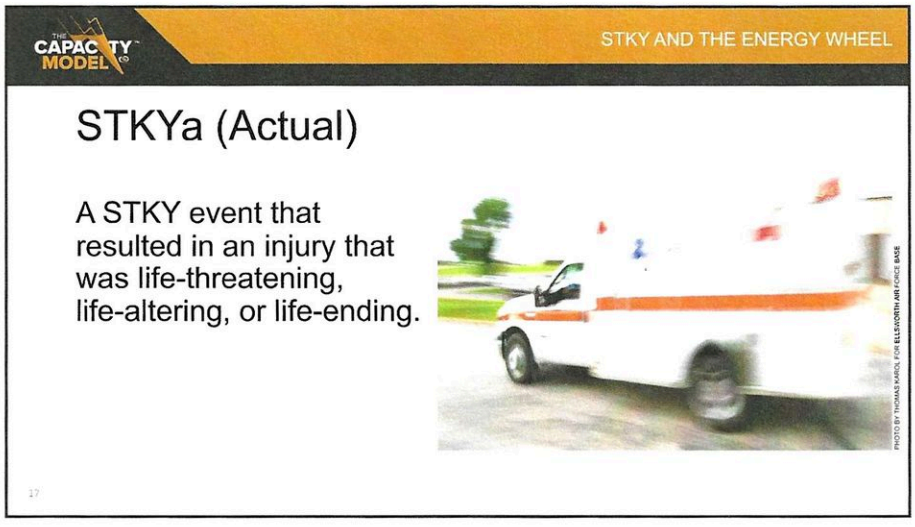
PRESENT THE FOLLOWING:

In the next few slides, we will discuss what makes an unwanted event a STKY event and discuss the classification of STKY events:

- STKYa: "a" stands for "actual"
- STKYp: "p" stands for "potential"
- STKY success.



- Review the definition of a STKY event.
- Discuss how it is different from an unwanted event such as a sprained ankle.
- Ask trainees: Can you think of an example of a STKY event in your work experience?
- Validate answers.



Slide aligns with page 11 of the trainee manual.

- Review definition on slide.
- Provide examples of each below

PRESENT THE FOLLOWING:

Life-threatening

- A state where death is imminent or probable if not immediately addressed by trained medical personnel, often with the aid of life-sustaining support. Examples of life-threatening injuries include:
  - Profuse bleeding
  - Damaged/blocked airway
  - Untreated shock or unnatural arrhythmia (irregular heartbeat due to unnatural causes like a shock or ingestion of a toxic substance).

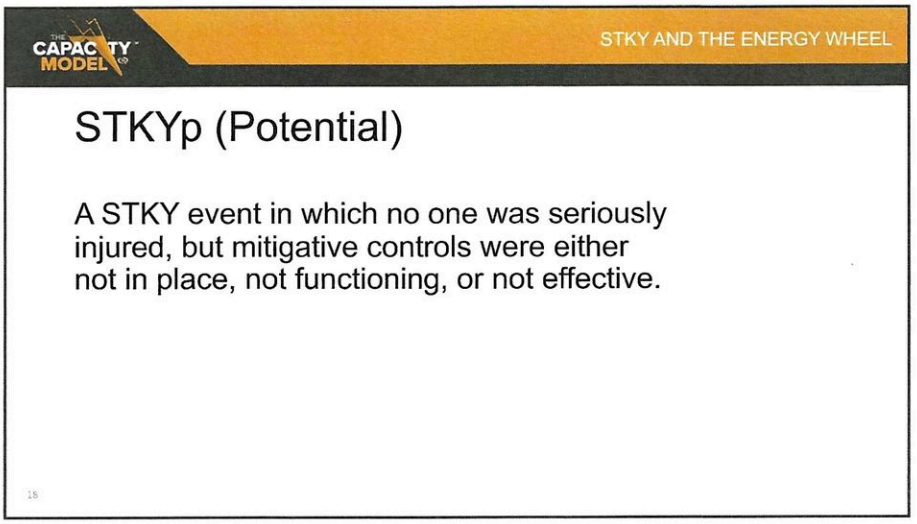
Life-altering

- A state of permanent or long-term impairment or loss of the use of an internal organ, bodily function, or body part. Examples of life-altering injuries include:
  - Amputations
  - Loss of vision/eye(s)
  - Compound fractures
  - Spinal cord injuries/paralysis
  - Life-altering head injuries
  - Severe burns (requiring grafting)
  - Disfigurement

Life-ending

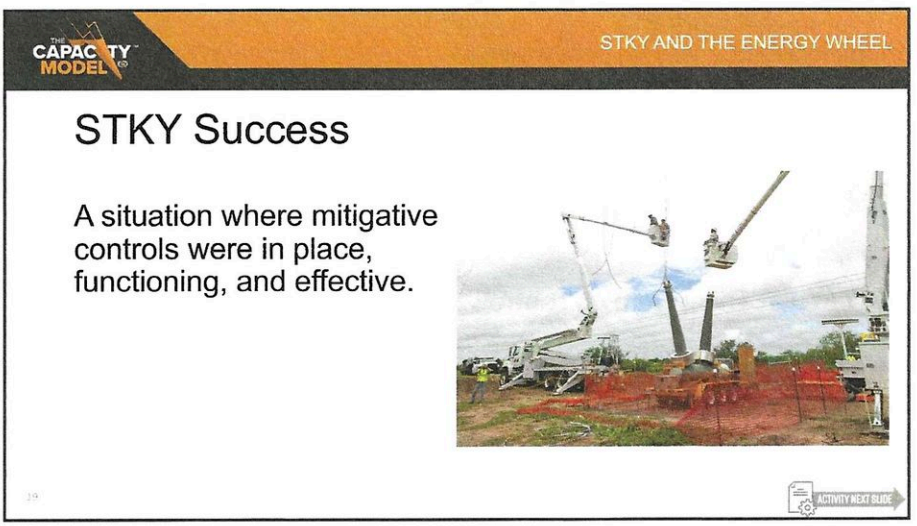
- An injury that results in the death of an employee.

Reinforce the guiding question:
Why do we need to be hyper-focused on STKY?



Slide aligns with page 12 of the trainee manual.

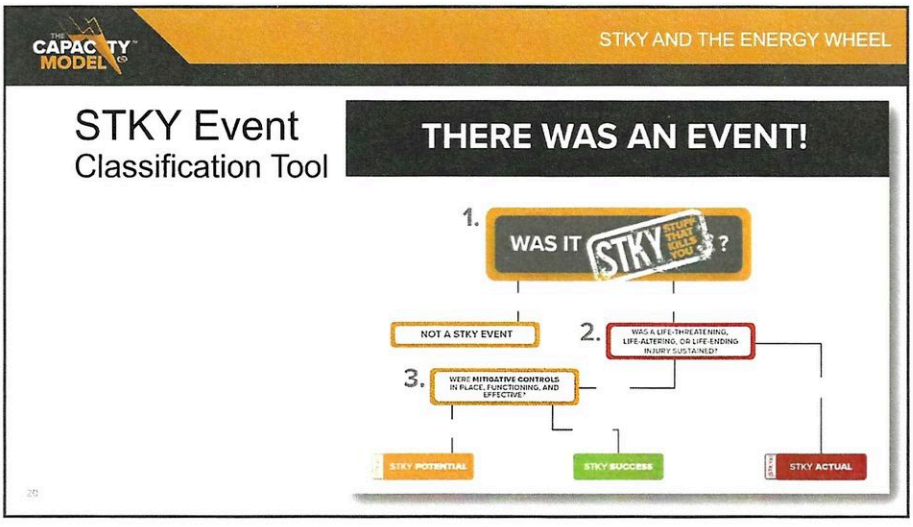
- Review definition: STKYp stands for STKY "potential." It is a STKY event in which no one was seriously injured, but mitigative controls were either not in place, not functioning, or not effective.
- Reinforce that the only thing protecting a crew from STKYa is luck.
- Direct trainees to page 12 of the manual to review the scenarios of STKYp.
- Ask trainees to explain in their own words, why each is a STKYp.



Slide aligns with page 13 of the trainee manual.

- Review definition: A situation where mitigative controls were in place, functioning, and effective
- These controls protected employees from an unwanted release of energy. The crew had capacity for failure that prevented a STKY event from becoming STKYa or STKYp.
- Direct trainees to page 13 of the manual to review the scenarios of STKY success.
- Ask trainees to explain in their own words, why each is a STKY success.

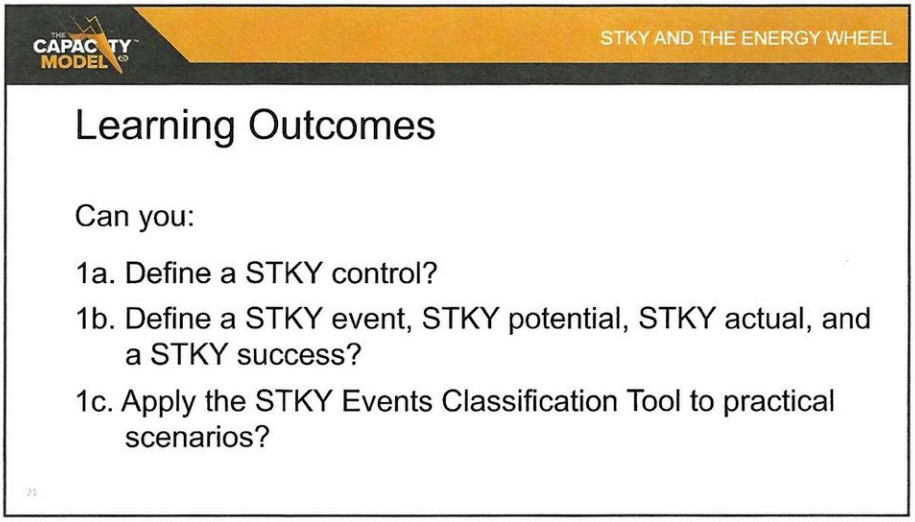
- Direct trainees to "A Good Catch" callout at the bottom of page 13. Discuss what a good catch is and review the examples.



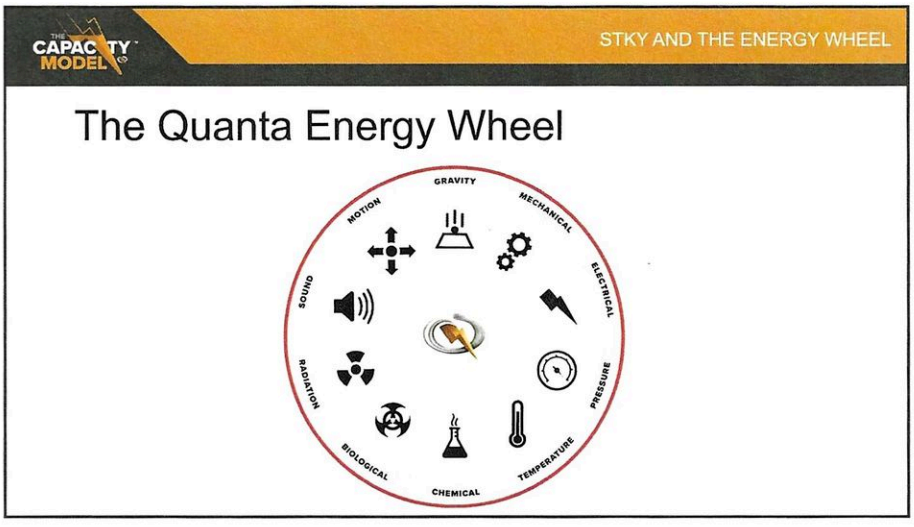
ACTIVITY:

- Direct trainees to the STKY Event Classification Tool and STKY Event Classification Tool Scenarios beginning on page 14.
- Walk through the STKY Event Classification Tool.
- Review instructions in the manual.
- Have trainees work in groups or individually to classify the remaining STKY scenarios using the classification tool.
- Review answers as a class. Ask trainees to explain why they chose their answer.
  1. STKY success: The event had the potential to be life-threatening, life-altering, or life-ending. It was a success because the the mitigative control, the loading dock restraint system, was in place, functioning, and effective.
  2. Not STKY: The event did not have the potential to be life-threatening, life-altering, or life-ending.
  3. STKY potential: The event had the potential to be life-threatening, life-altering, or life-ending. There were no mitigative controls in place. The crewmember was lucky she was not injured.
  4. STKY actual: There were no controls in place. The worker suffered a life-altering injury.
  5. Not STKY: No unwanted energy was released (gravity). The worker took the damaged positioning strap out of service. **This was a good catch**.

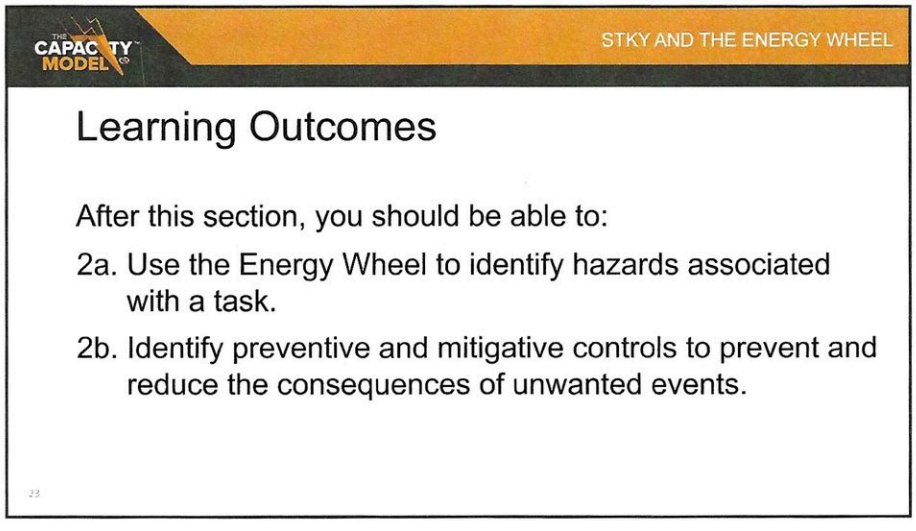
Optional activity:
Engage with trainees and ask them to think of a possible STKY example they've encountered. Use that scenario with the classification tool to determine the event's classification



- Review the learning outcomes.
- Ask trainees to:
  - Define a STKY control.
  - Explain the difference between a STKYa and a STKYp.
  - Explain what a STKY success is.
- Ask for volunteers to answer the questions.
- Ask trainees if they have unanswered questions or points of confusion about STKY controls or STKY events.

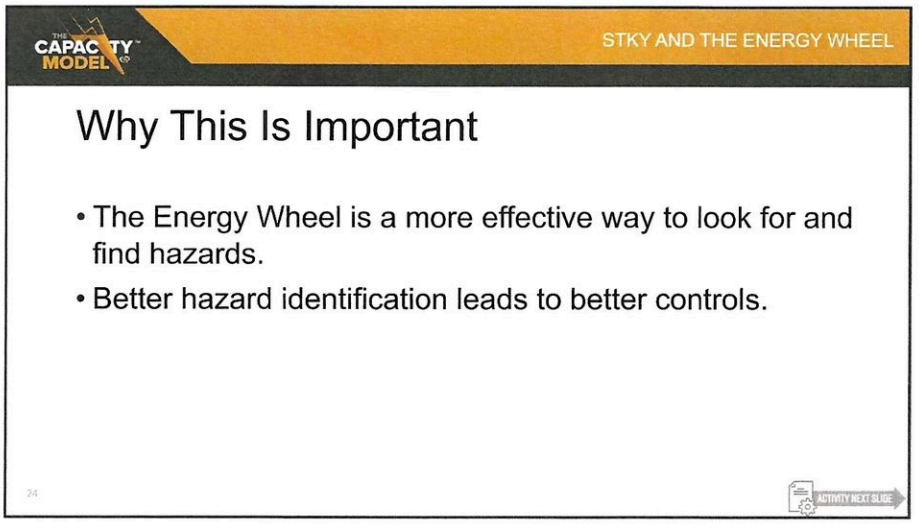


- Ask trainees: "What do you know about the Energy Wheel?" or "What is your experience with the Energy Wheel?"
- Ask for 1-2 volunteers to share their responses.
- Validate responses.
- Discuss the purpose of this session, for example:
  - The Energy Wheel is a tool that helps crews visualize and identify more hazardous energies.
  - It does not replace the job briefing, but it is a tool to use with the job briefing.
  - This session guides trainees through the process for holding STKY discussions, using the Energy Wheel to identify hazardous energies, identify unwanted events from those energies, and put preventive and mitigative controls in place to prevent and reduce the consequences of the unwanted events.



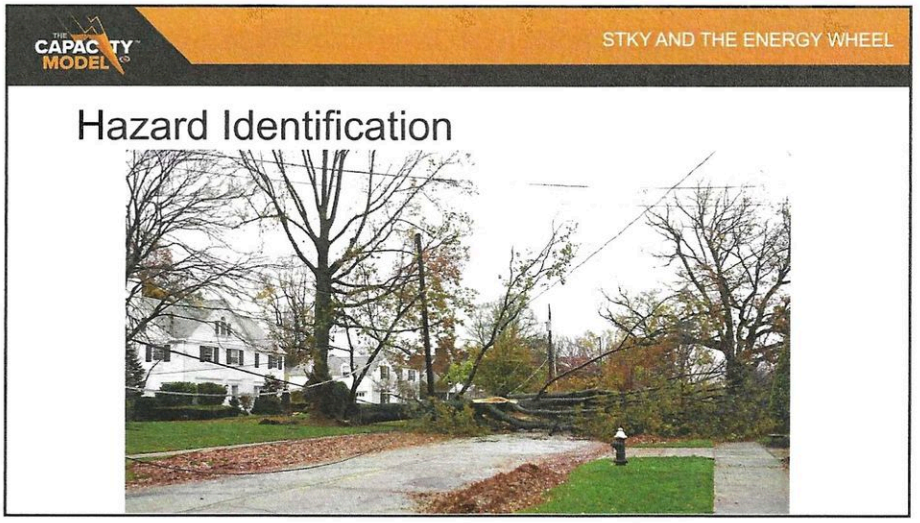
Slide aligns with page 17 of the trainee manual.

- Review learning outcomes.
- Read or ask for volunteers to read the learning outcomes.
- Write the learning outcomes on a whiteboard or post them in a visible area of the room to refer to during the training.



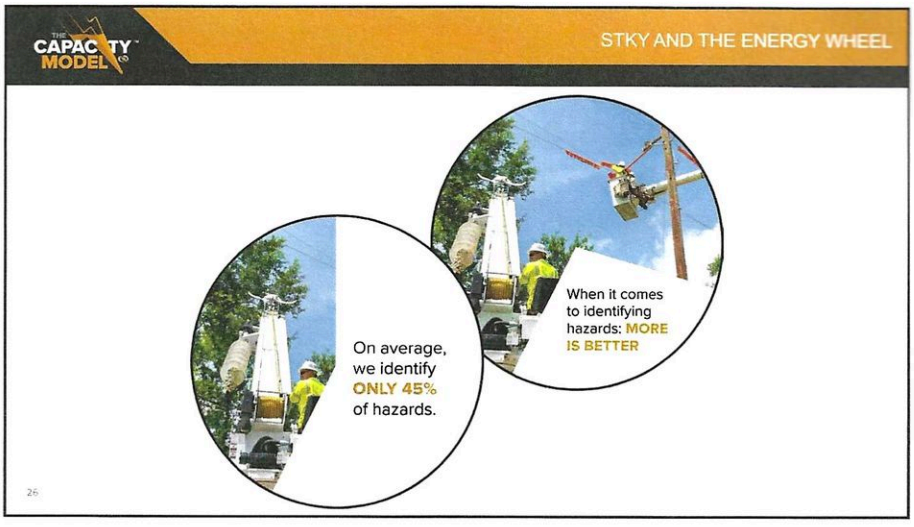
- Expand on each bullet item.

Reinforce the guiding question:
Why do we need to be able to identify all of the Energy Wheel hazards?



ACTIVITY:

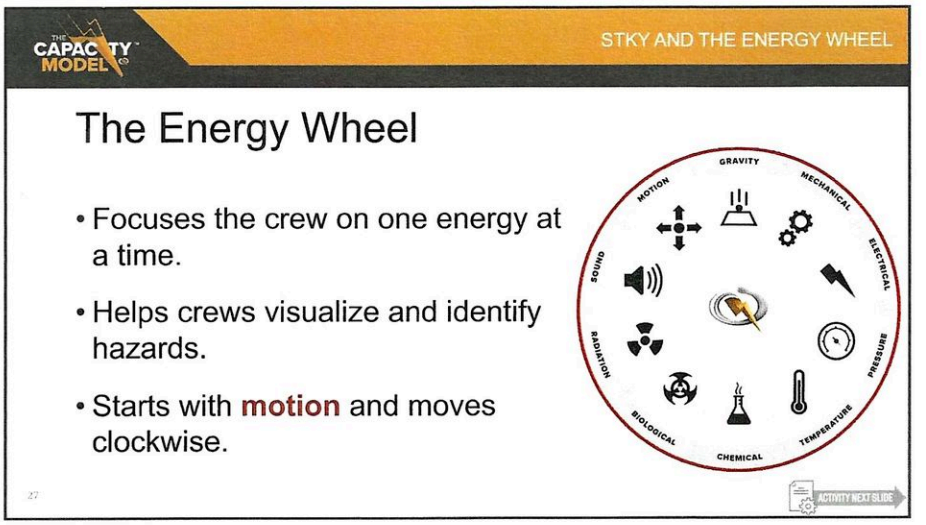
- Purpose: Trainees will identify hazards without using the Energy Wheel. Then, using the Energy Wheel, they can visualize how many more hazards they were able to identify.
- Present the storm restoration scenario to trainees.
- As a class, ask trainees to identify hazards in the photo without the Energy Wheel.
- Identify a hazard to get the discussion started.
- Capture these hazards on a whiteboard or flip chart.
- After you present the Energy Wheel and the energies, you will revisit the photo to identify hazards using the Energy Wheel.



Slide aligns with page 18 of the trainee manual.

- Discuss the left circle of the graphic: 45% is the average percentage of hazards that workers can identify on their job sites. That means more than half of all hazards typically go unrecognized.
- When evaluating a job and job site for hazards, it's a challenge to account for every possible hazard and keep track of them throughout the workday.
- Humans have a lot of competing demands, distractions, and factors that can get in the way of effective hazard identification.
- Research shows hazard identification can jump from 45% to over 70% when an energy-based approach is used to focus a crew's attention (right graphic). (Albert et al, 2017, Albert, Alex, 2013)

Reinforce the guiding question:
Why do we need to be able to identify all of the Energy Wheel hazards?



Slide aligns with pages 22-23 of the trainee manual.

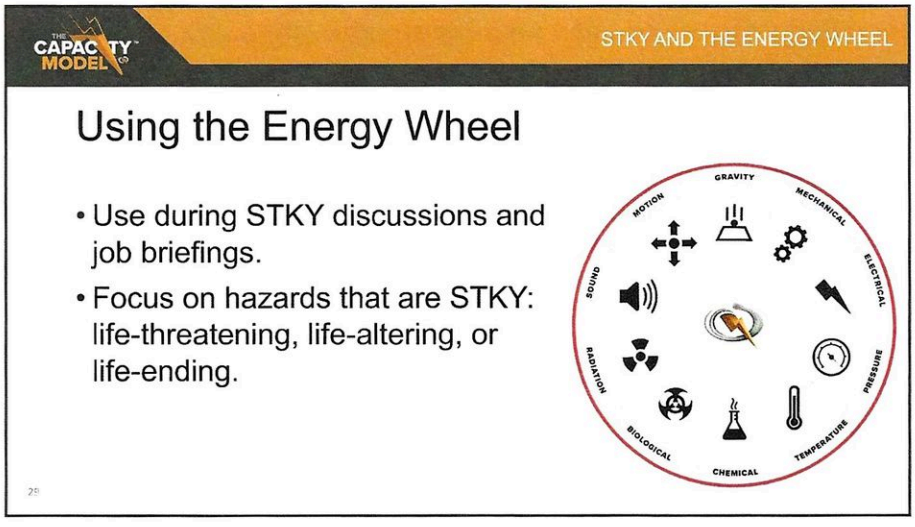
- Direct trainees to pages 22 to follow along.
- Discuss why the EnergyWheel is an important tool:
  - The Energy Wheel is a tool that takes an energy-based approach to identifying hazards. It helps crews visualize and identify hazardous energies by focusing on one energy at a time.
  - This is important on large jobs with multiple tasks that see multiple energies.
  - It is important to use on small, seemingly routine jobs.

- Discuss the format of the Energy Wheel:
  - The most easily recognized energy hazards are at the top of the Energy Wheel, beginning with motion.
  - They are strategically placed in this order to show the most commonly identified to the least identified energies as you move clockwise around the wheel, ending with sound.

- Direct trainees to page 23, Table 2.1, to review the elements of Energy Wheel.
  - Review the shoutables.
  - Provide an example of each.

ACTIVITY:

- Revisit the photo. Direct trainees to page 22 for a graphic of the Energy Wheel.
- Capture the additional hazards on the whiteboard or flip chart. Use a different color marker or use a separate column to add to the list.
- Emphasize the additional hazards identified by using the Energy Wheel.

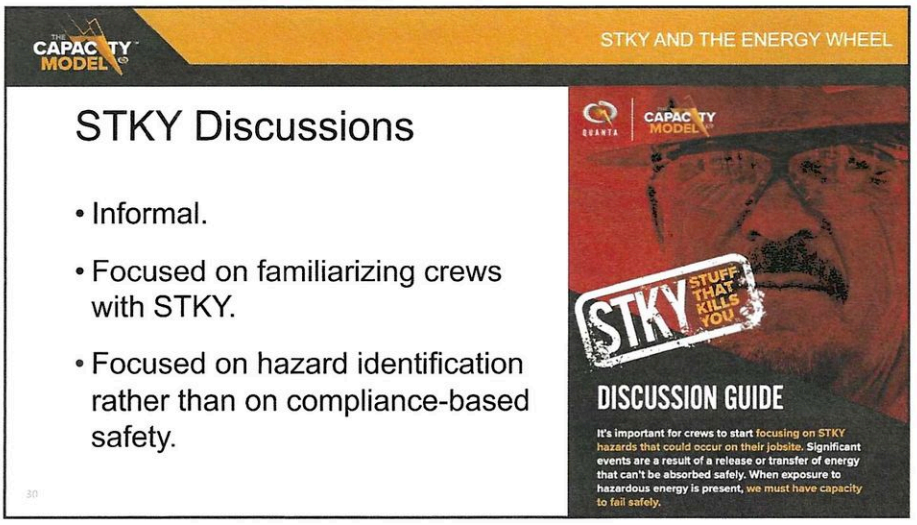


Slide aligns with page 24 of the trainee manual.

PRESENT THE FOLLOWING:

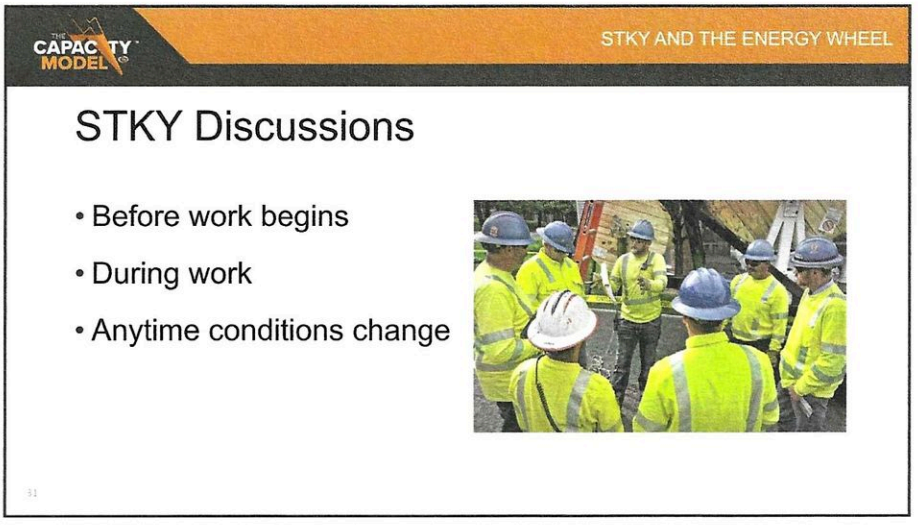
Explain how to use the Energy Wheel:

- The Energy Wheel is used to guide crews through hazard identification during job briefings and STKY discussions.
- While the Energy Wheel can be used to identify all hazards, crews want to focus on STKY: life-threatening, life-altering, or life-ending events.
- The conditions that cause a sprained ankle are rarely the same as the ones that cause a fatality. Workers can recover from an ankle sprain. They can't recover from a life-ending event.
- As crews complete the job briefing and discuss the tasks, recognize that the purpose of the Energy Wheel is not a hunt to identify a hazard for every energy. **Not all energies are going to be present for every job on every job site.**



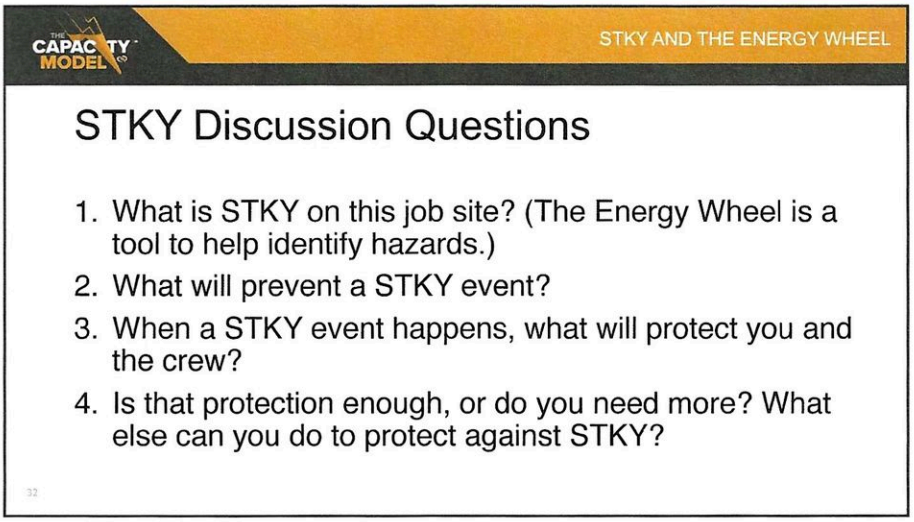
Slide aligns with page 25 of the trainee manual.

- Reference the STKY Discussion Guide they received at the beginning of class.
- Discuss the elements of a STKY discussion. Expand on bullet items on slide. For example:
  - Use informal discussions to help workers develop their understanding of STKY.
  - Create a safe space where workers feel comfortable to make mistakes while they learn about STKY discussions. The discussions should be open and designed to put crews at ease. Help them to feel comfortable sharing their thoughts.
  - STKY discussions are focused on identifying hazards at work and hazards associated with a job or task. They are not meant to hold compliance-based training.



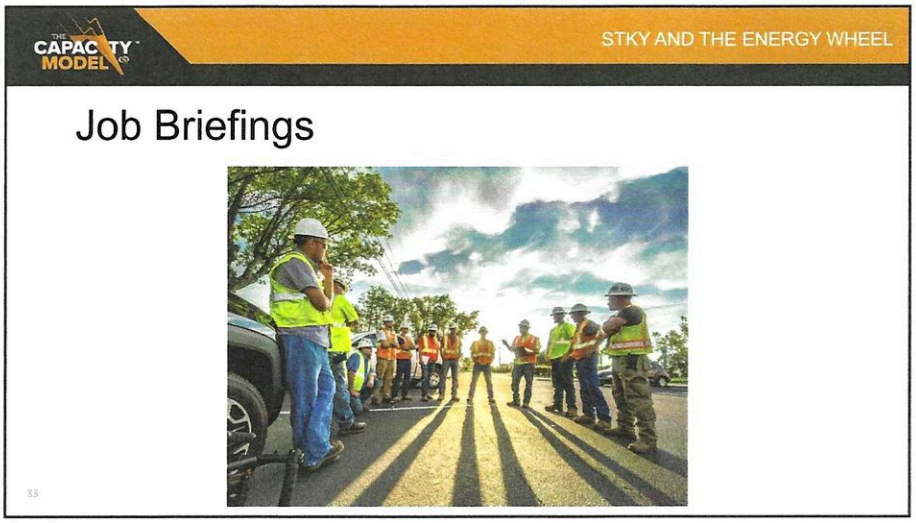
Discuss when to hold STKY discussion. Expand on bullet items. For example:

- During the job briefing.
- When the task changes.
- If there is a change in equipment.
- If the environment or weather changes.



Walk through the four questions, explaining each:

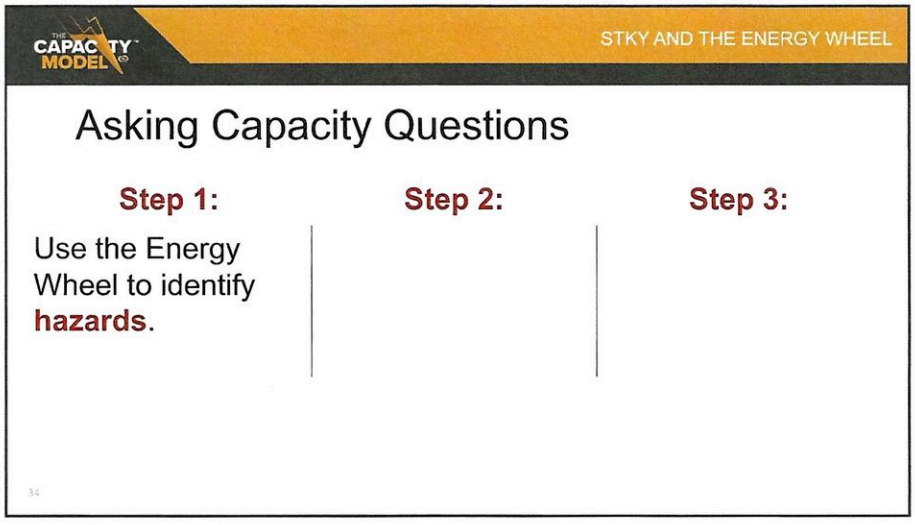
1. Use the Energy Wheel to identify what is STKY. Walk through each energy.
   - Reinforce that the Energy Wheel is a tool crews use to look for energies on a job site and in a job.
2. What will prevent a STKY event is the question you ask to identify preventive controls.
3. When a STKY event happens, what will protect you and the crew? The question focuses attention on mitigative controls.
   - Reinforce that prevention is not enough. Work must be planned for when—not if—failure occurs.
4. Is that protection enough or do you need more? The question prompts the crew to find additional layers of controls.
   - Reinforce that more layers of controls make safer working conditions.



Slide aligns with page 27 of the trainee manual.

- Ask trainees: What are the components of your company's job briefing?
- Ask for 1-2 volunteers to share.
- Validate answers. Reinforce hazard identification if appropriate.

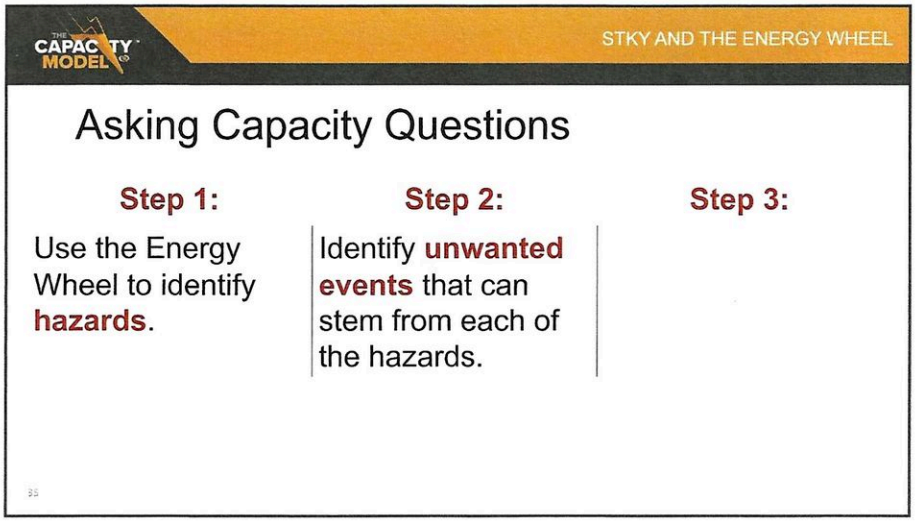
- Explain that the Energy Wheel does not replace any of the components of a job briefing.
- It is a tool that can be used during the hazard identification portion of a job briefing.
- The Energy Wheel's structured approach will help every employee notice hazards they may otherwise miss.



Slide aligns with page 29 of the trainee manual.

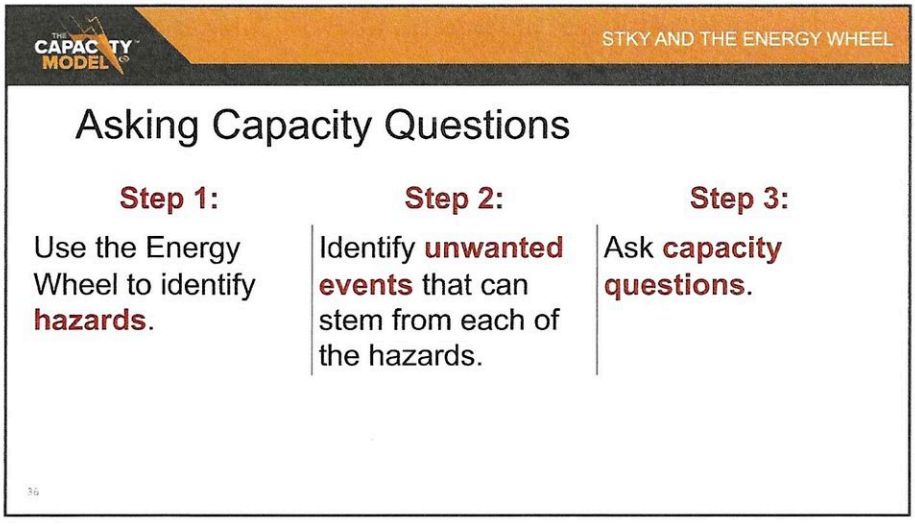
- Explain that leaders and their crews use the Energy Wheel during the hazard identification component of the job briefing.
- Once hazards are identified, leaders and their crews work through the process to put preventive and mitigative controls in place.
- Walk trainees through step 1:
  - Walk through each energy on the Energy Wheel to identify the hazards that are present on the job site and for each task of the job.
  - Provide an example: If you are using a backhoe to dig a trench, the boom and bucket are a motion hazard.

- Ask trainees: Do we need to be able to identify a hazard for all energies on the Energy Wheel?
- Answer: No. Not all energies are going to be present on every job site or task.



Walk trainees through step 2:

- For every hazard crews identify, they must then identify unwanted events that can stem from that hazard.
- Review the definition of an unwanted event.
- Expand on definition: It is the bad stuff that can happen because of a hazard.
- Provide an example: If the boom and bucket of a backhoe create a motion hazard, an unwanted event is a worker getting struck by the boom or bucket.

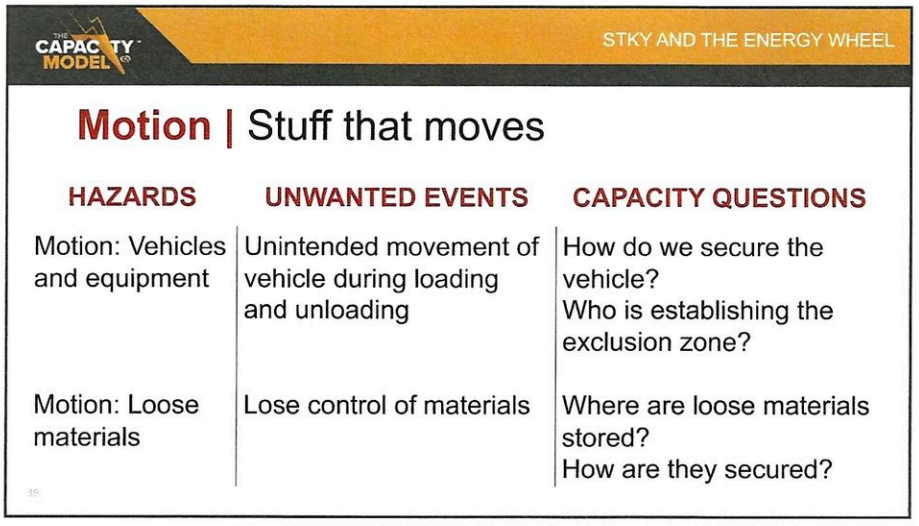


- Walk trainees through step 3:
  - For each unwanted event, crews must ask capacity questions to identify preventive and mitigative controls to prevent and reduce the consequences of the unwanted event.
  - Review the definition of capacity questions.
  - Provide an example: If the unwanted event is a worker getting struck by the boom or bucket of a backhoe, crews need to ask capacity questions such as, "Who is the spotter assigned to the equipment?" and "How is the swing radius marked or barricaded?"

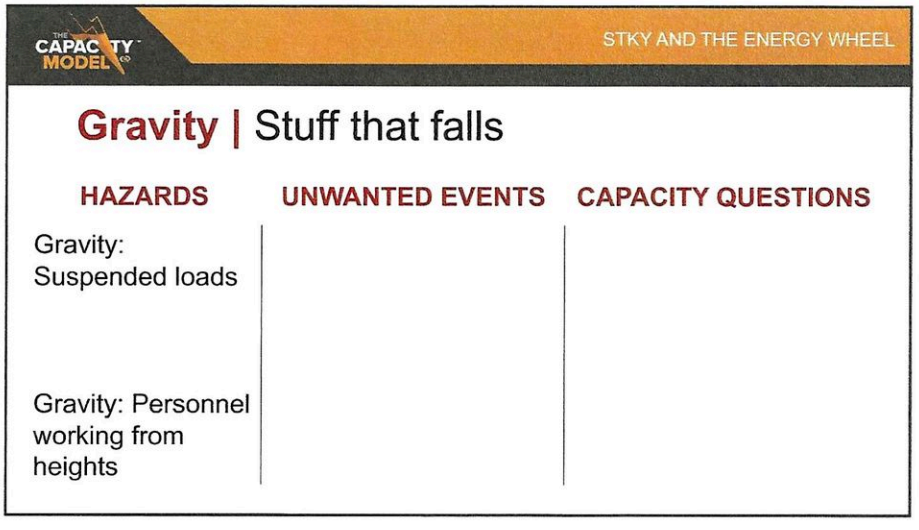
- Explain you will work through this process for each of the energies on the following slides.
- Direct trainees to pages 30-39 if they want to see sample capacity questions for each of the energies.

- Discuss each motion hazard.
- The next slide displays an unwanted event for each hazard.

- Walk through the unwanted event for the hazard.
- Explain this is not a complete list of unwanted events.
- The next slide displays potential capacity questions for the unwanted events.



- Walk through the sample capacity questions.
- Explain this is not a complete list of capacity questions.
- Reinforce that capacity questions are meant to identify preventive and mitigative controls.

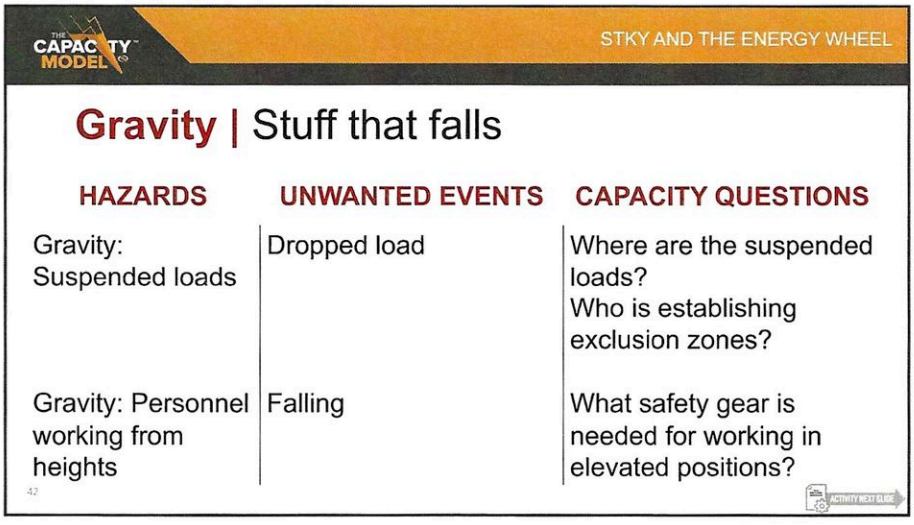


ACTIVITY:

- Complete the gravity hazards as a class.
- Walk through each hazard.
- Ask for volunteers to identify unwanted events that can stem from each hazard.
- Record the hazards and unwanted events on a whiteboard or flip chart before moving to the next slide.

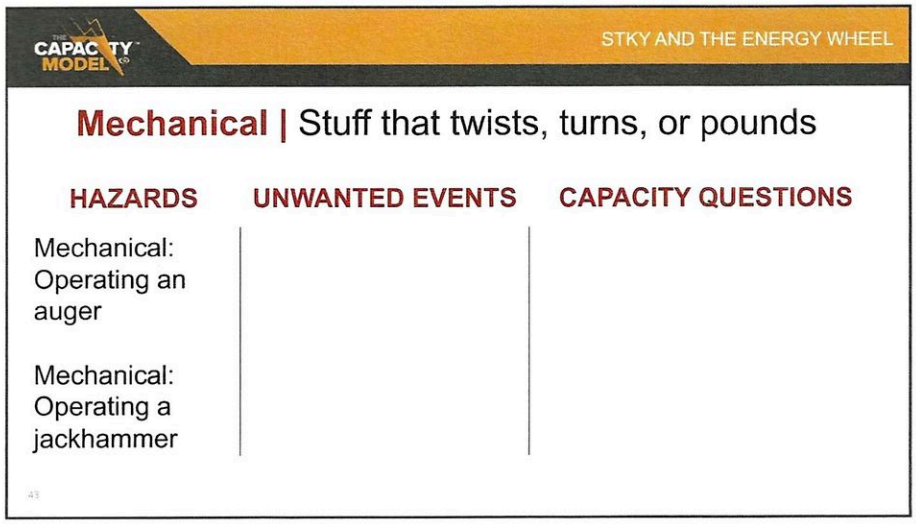
ACTIVITY:

- Walk through the example unwanted events on the slide.
- Ask trainees: For the unwanted events we identified, what are some capacity questions we can ask to help us identify preventive and mitigative controls for the unwanted event?
- Record answers on a whiteboard or flip chart before moving to the next slide.



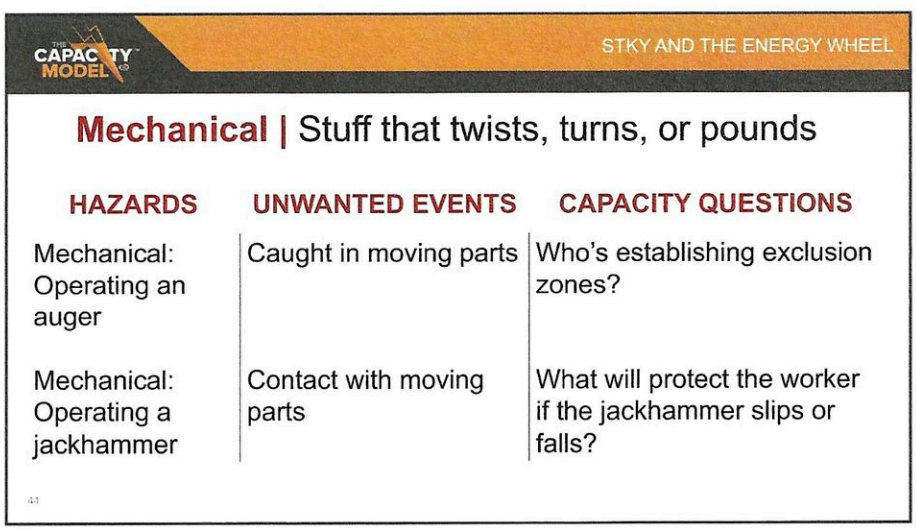
ACTIVITY:

- Review the capacity questions on the slide.
- Explain this is not a complete list of unwanted events or capacity questions.



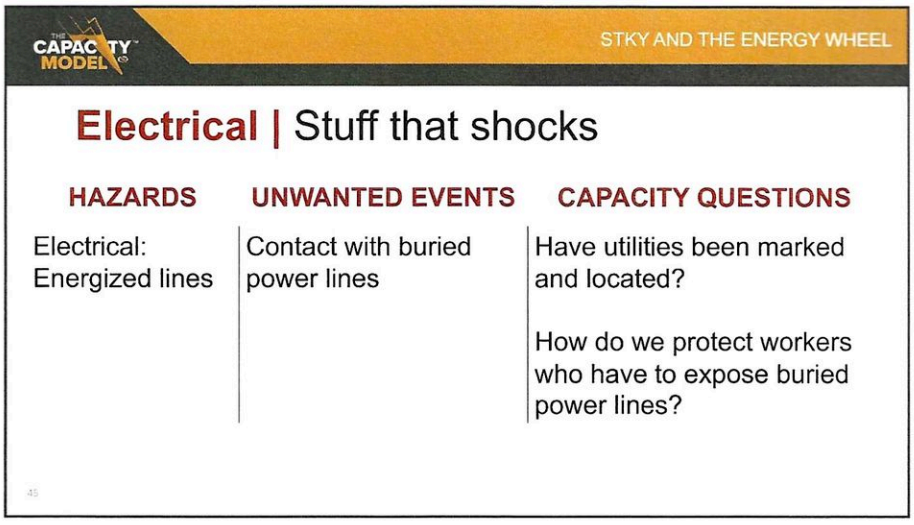
ACTIVITY:

- Review each of the mechanical hazards.
- Direct trainees to work with a partner or small group to identify an unwanted event for each hazard and capacity questions to help identify controls.
- Provide five minutes to complete the activity.
- Ask for volunteers to share their responses.
- Validate answers before moving to the next slide.

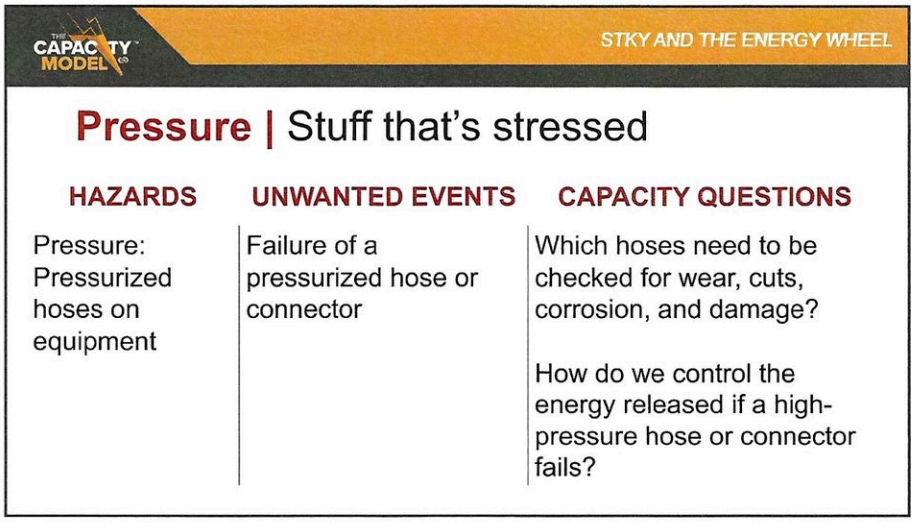


ACTIVITY:

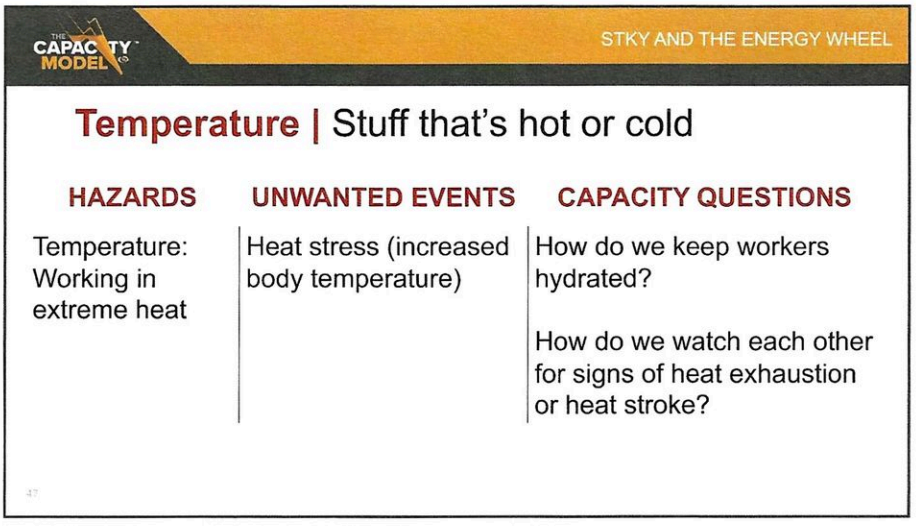
- Review the sample unwanted events and capacity questions for each hazard.
- Explain this is not a complete list of unwanted events or capacity questions.



- Explain for the remaining Energy Wheel energies, you will walk through an example of each.
- Walk through the example of an electrical hazard.
- Confirm understanding of the energy before moving on.

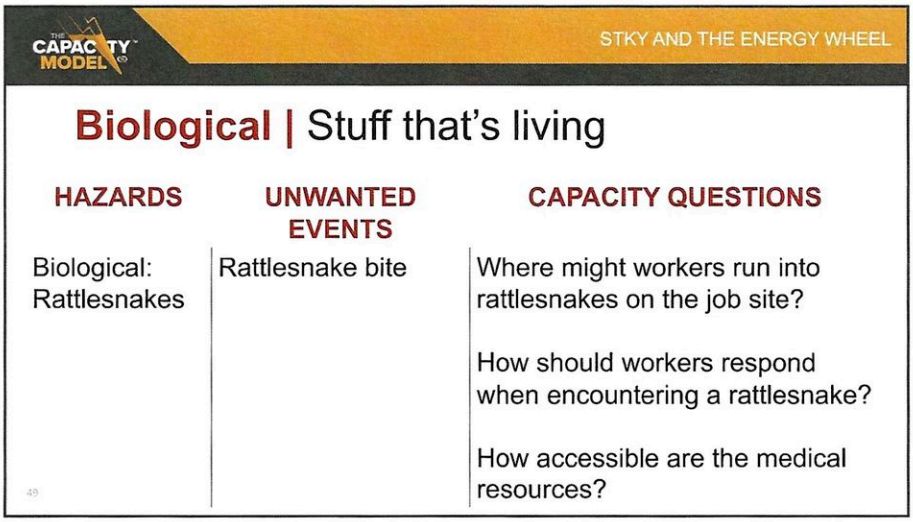


- Walk through the example of a pressure hazard.
- Reinforce that pressure does not include mental pressure to get a job done on time.
- Confirm understanding of the energy before moving on.

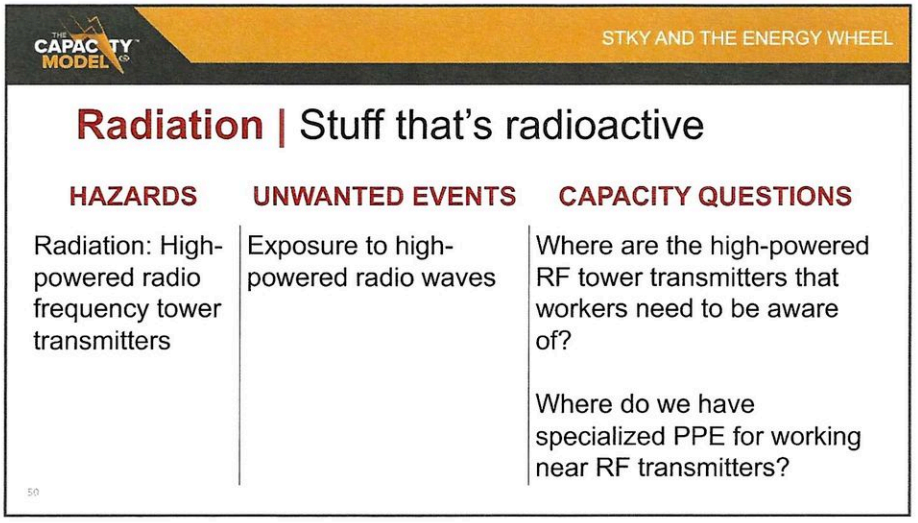


- Walk through the example of a temperature hazard.
- Confirm understanding of the energy before moving on.

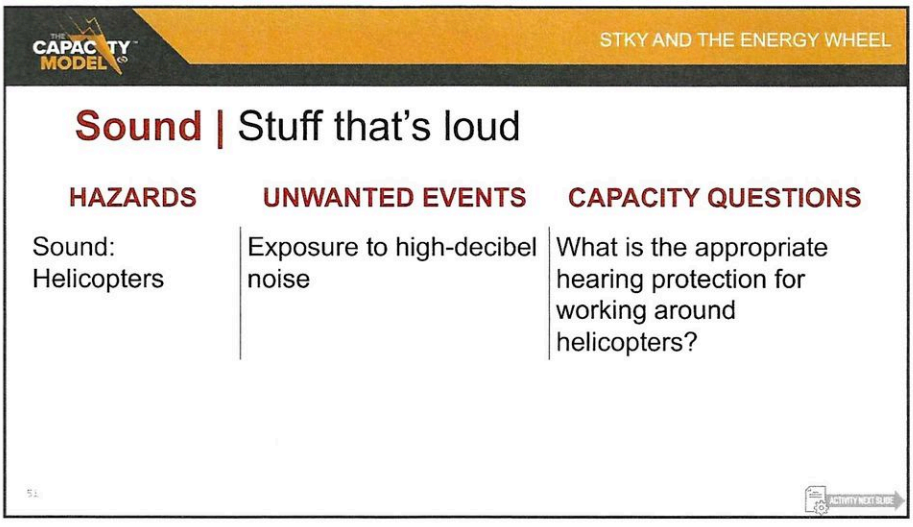
- Walk through the example of a chemical hazard.
- Confirm understanding of the energy before moving on.



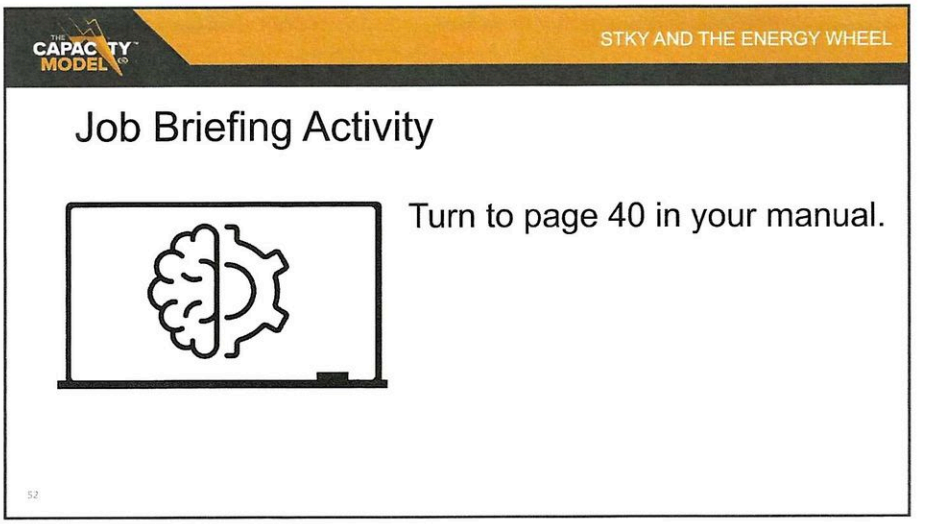
- Walk through the example of a biological hazard.
- Confirm understanding of the energy before moving on.



- Walk through the example of a radiation hazard.
- Confirm understanding of the energy before moving on.



- Ask trainees: Why do sound hazards fall into the category of a STKY hazard?
- Possible answers:
  - Permanent damage to hearing (life-altering event).
  - Loud sounds can make other warning sounds (like an alarm) inaudible, leading to a STKY event.
  - A noisy environment can affect communication between crewmembers, leading to a STKY event.
- Walk through the example of a sound hazard.
- Discuss how exposure to high-decibel noise whether acute and immediate or less-severe and gradual are both STKY:
  - Explain that unprotected exposure to extremely high-decibel noises like detonated dynamite have an immediate impact. The worker will experience immediate permanent hearing loss. It is STKY because permanent hearing loss is a life-altering event.
  - Explain that noise hazards such as unprotected exposure to helicopters and jackhammers can approach the decibel threshold for causing immediate permanent damage, but permanent hearing loss will happen when unprotected exposure occurs over a period of time. This is STKY because gradual permanent hearing loss is still a life-altering event.



ACTIVITY:

- Purpose: Trainees will complete the job briefing activity to practice using the Energy Wheel to identify hazards, identify unwanted events, and ask capacity questions to identify preventive and mitigative controls for the unwanted events.
- The job briefing activity in the trainee manual includes four scenarios. Two power scenarios are on pages 40 and 42. Two telecom scenarios are on pages 44 and 46. Choose the scenarios based on your audience.
- The first scenario trainees will complete as a class. In the second scenario, students will work with partners or small groups to complete.
- See pages 42 and 43 of this manual for the power scenarios and an example of a completed task.
- See pages 44 and 45 of this manual for the telecommunications scenarios and an example of a completed task.

Complete the first scenario as a class:

- Recreate the boxes of the activity on a whiteboard or flip chart.
- Review the directions with the class.
- Work through each step of the activity.

Trainees will complete the second scenario with a partner or small group:

- Provide 15 minutes to complete the activity.
- Walk around to check in and answer questions.
- Ask for volunteers to walk through a task and the hazard, unwanted event, and preventive and mitigative controls for the unwanted event.

JOB SCENARIO: You are performing live-line work spreading wire for a reconductor job.

1. Task: Lay out wire and apply mechanical jumpers.

4. Preventive Controls (Prevent the unwanted event):

- Installation of rubber goods/cover-up.
- Designated observer to make sure mechanical jumpers (macs) are installed on the correct phase.

2. Hazard/Energy: Electrical

3. Unwanted Event: Arc flash

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

- PPE: rubber gloves and sleeves, FR clothing

1. Task:

4. Preventive Controls (Prevent the unwanted event):

2. Hazard/Energy:

3. Unwanted Event:

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

Use the extra space on the concept maps to record more examples in case trainees need additional help.

JOB SCENARIO: You are completing a pole changeout project to replace several poles that run alongside a four-lane road and the backyard fences of private homes.

1. Task: Block a lane of traffic to set up the work zone.

4. Preventive Controls (Prevent the unwanted event):

- Set up worksite protection: traffic cones, lane redirect, flagging service.

2. Hazard/Energy: Motion

3. Unwanted Event: Vehicle entering work zone

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

- Set up blocker vehicle.
- Set up water-filled barrels or another type of barrier.

1. Task:

4. Preventive Controls (Prevent the unwanted event):

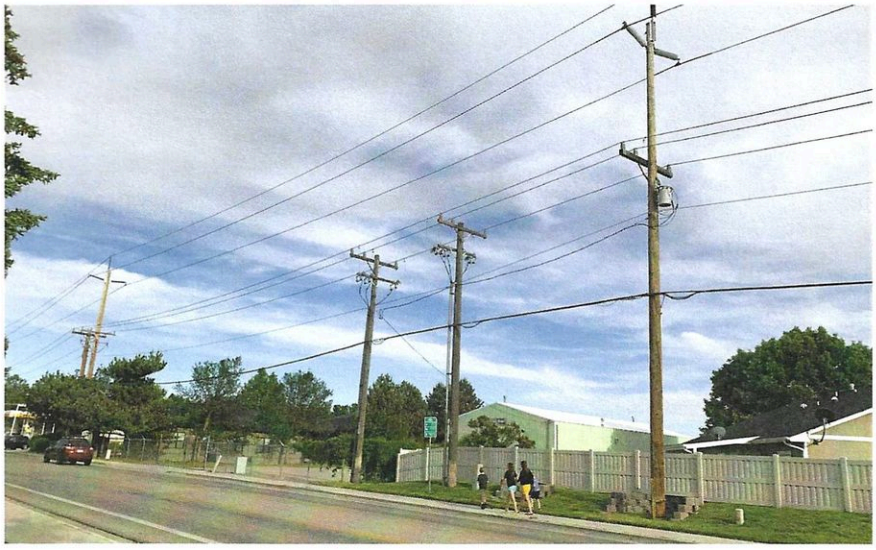
2. Hazard/Energy:

3. Unwanted Event:

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

Use the extra space on the concept maps to record more examples in case trainees need additional help.

JOB SCENARIO: Your job is to lower a fiber splice using an aerial lift. You will need to block one lane of traffic to lower the splice.



1. Task: Block a lane of traffic to set up the work zone.

4. Preventive Controls (Prevent the unwanted event):

- Set up worksite protection: traffic cones, lane redirect, flagging service

2. Hazard/Energy: Motion

3. Unwanted Event: Vehicle entering work zone

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

- Set up blocker vehicle.
- Set up water-filled barrels or another type of barrier.

1. Task:

4. Preventive Controls (Prevent the unwanted event):

2. Hazard/Energy:

3. Unwanted Event:

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

Use the extra space on the concept maps to record more examples in case trainees need additional help.

JOB SCENARIO: You are installing new equipment at the top of a cellular tower. You will set up rigging to send the new equipment up.

1. Task: Set up rigging to send up the replacement equipment.

4. Preventive Controls (Prevent the unwanted event):
- Consult load charts.
- Inspect and verify rigging material.

2. Hazard/Energy: Gravity

3. Unwanted Event: Dropped load

5. Mitigative Controls (Mitigate the consequences of the unwanted event):
- Set up exclusion zone.
- Use three-way communication.
- Assign a spotter.

1. Task:

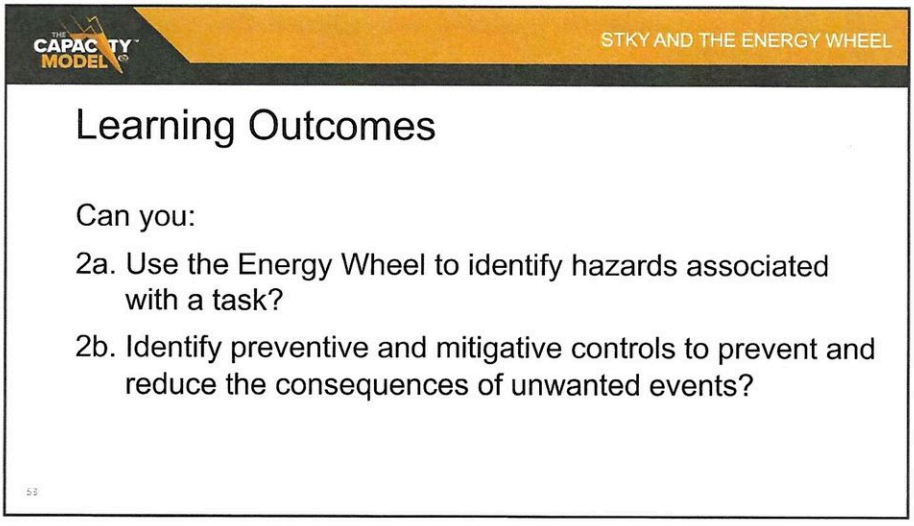
4. Preventive Controls (Prevent the unwanted event):

2. Hazard/Energy:

3. Unwanted Event:

5. Mitigative Controls (Mitigate the consequences of the unwanted event):

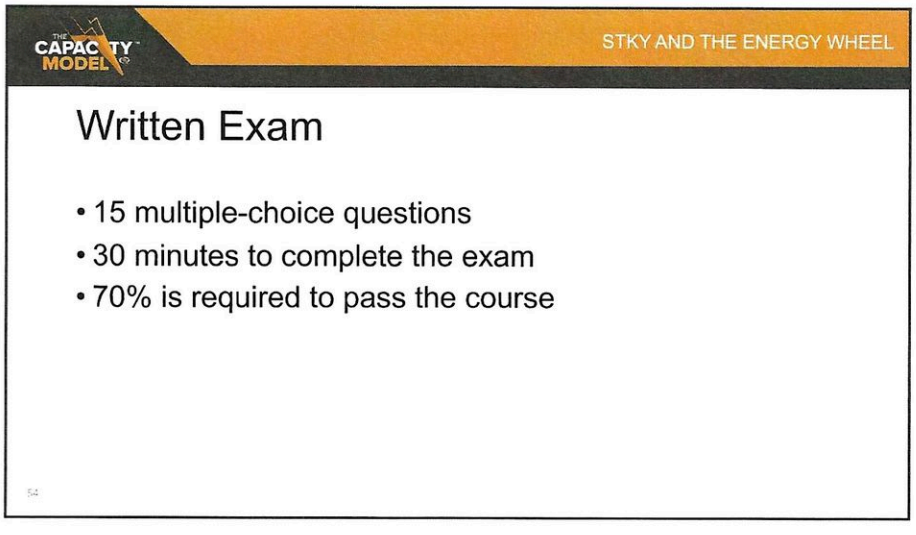
Use the extra space on the concept maps to record more examples in case trainees need additional help.



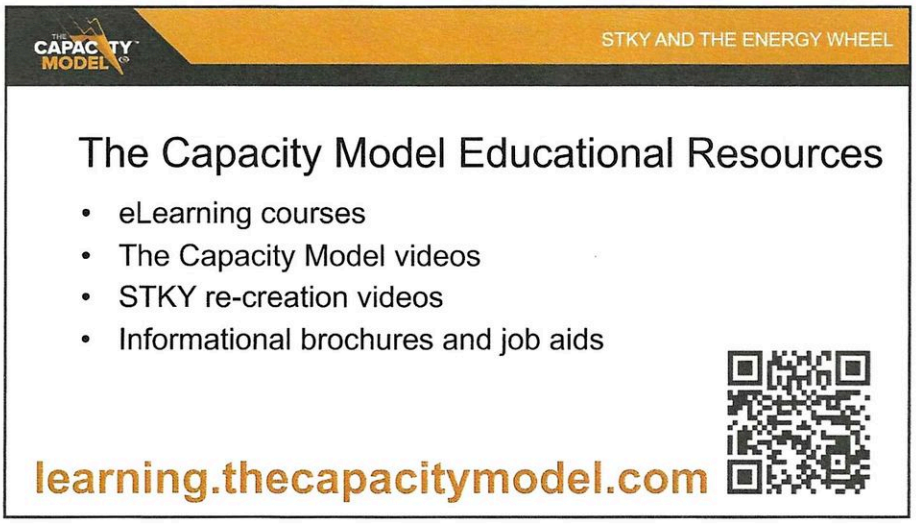
- Review the learning outcomes. Discuss each with the trainees. For example:
  - Ask for volunteers to summarize a learning outcome.
  - Ask volunteers to share one thing they learned related to a learning outcome.
- Clarify anything trainees do not understand.
  - Ask trainees: What are your questions?

After asking for questions, count to seven in your head before moving on.

See Appendix D for other Formative Assessment ideas.



- Review bullet items for the written exam.



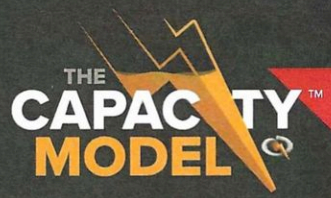
- Direct trainees to learning.thecapacitymodel.com or the QR code to view and download TCM educational resources.
- Explain the site is open. They do not need to register or log in.
- Review the resources:
  - eLearning courses such as STKY Controls for Crews and Learning Teams.
  - TCM videos like those viewed in this course.
  - STKY re-creation videos are re-creations of real STKY incidents from Quanta operating companies.
  - Informational brochures and job aids such as the STKY discussion questions are available to download.

EXAM GUIDE

- Pass the exams out to the trainees.
- After everyone finishes, instruct trainees to swap exams with a partner.
- Review and grade the exams as a class. Discuss each question and answer.
- Collect exams.

WRAP-UP

- Ask trainees to complete the instructor evaluation form.



APPENDICES

APPENDIX A—COURSE DELIVERY TIPS

BEFORE DELIVERY

- Review the instructor manual to stay fresh on the information.
- Star your favorite application activity and formative assessment ideas in the appendix so you can use them in your lesson.
- Make sure you have all your materials ready and accessible.
- Set the room up in a way that encourages discussion and participation in activities. For example, having the students sit around tables instead of in rows.
- Use the tools available to you. Don't have access to mini whiteboards? What about a page protector with a piece of printer paper in it?

DURING DELIVERY

- Build a positive, professional classroom culture.
  - Set clear expectations.
  - Encourage all questions and comments – this is a safe place. Disagreements can be made in a professional and respectful way.
  - Complimenting can go a long way.
  - Smile!
  - Provide breaks and monitor student body language to decipher when to use them. Consider designating a student to help you know when it is a good time for a break.
- Don't just read the learning outcomes – decipher them for the student. What is in it for them?
  - Write the learning outcomes on the whiteboard or on a wall chart to reference throughout the lesson.
- Start your lessons by motivating the students and re-motivate throughout your delivery.
  - A quick and easy way to monitor engagement is to walk around the room during the lesson.
  - Making eye contact is an excellent way to motivate students to stay engaged.
- Spend plenty of time on the application activities—these are the meat and potatoes of the lesson!
- Plan three to five extra application activities to use in the lesson. If you have extra time incorporate them. If not, it's no big deal!
- Involve students in the presentation – you should be talking less than they are!
  - Use students' experiences to help each other learn. Your job is to facilitate and guide the discussions around the topics.
- Evaluate throughout the lesson to check student understanding before moving on to a new concept.

AFTER DELIVERY

- Write notes in your instructor manual to improve your delivery after each lesson.
  - Make note of things you would like to try.
  - Make note of things that went well.
  - Make note of things that were challenging.

APPENDIX B—RECOMMENDED RESOURCES AND MATERIALS FOR ACTIVITIES AND FORMATIVE ASSESSMENTS

Each activity or assessment in Appendices B, C, and D has accompanying materials and/or other components to prepare before class starts.

The following is a list of recommended materials to have on hand so you can be prepared for most of the activities. You should make sure you have enough of each item for each student to have a set.

NAME	DESCRIPTION
ABC Cards	All letters of the alphabet printed on cardstock and cut out. *Tip: Single-hole punch a corner and stick them on a binder ring so that they don't get lost or mixed up easily.*
Dry erase markers	Thin tip dry erase markers in assorted colors.
Chart Paper	Large sheets of unlined paper that can be attached to the wall by either adhesive on the back or tape.
Envelopes	Envelopes for activities that require sorting or containing small cutout items.
Lined Paper	College ruled or wide ruled lined paper
Magnets	A baggy of small magnets to use with magnetic whiteboards.
Markers	Markers in assorted colors.
Pens/Pencils	Black or blue pens and number two pencils.
Printer Paper	White unlined paper.
Tape	Masking or painters tape so that it will stick to most surfaces without damaging anything
Whiteboard erasers	Individual small whiteboard erasers, or new socks or wash cloths for increased portability
Whiteboards	Individual student whiteboards, or a piece of printer paper in a page protector for increased portability.

APPENDIX C—APPLICATION ACTIVITY MENU

NAME	DESCRIPTION / STEPS	MATERIALS / PREPARATION
3-2-1	1. Ask students to list three things. 2. Ask students to list two things. 3. Ask students to list one thing. *Example: Write the three elements of The Capacity Model™. Write two scenarios depicting STKYa events. Write one question you still have about the topic.*	• Paper • Pens ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers
ABC Words/Phrases	1. Give each student, pair, or group a letter. 2. Have students find a word or phrase from the lesson that starts with that letter. 3. Have students share their words/phrases.	*Optional:* • ABC Cards
Acrostic	1. Have students write a lesson-related word vertically. 2. Have them use each letter in the word to write a word/phrase that relates to the topic. 3. Have students volunteer to share.	• Paper • Pens ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers
Baseball	1. Have students divide into two evenly-sized groups. 2. Have two players from each group come up to the whiteboard. 3. Provide a question. 4. The first player to write the correct answer gets a point for their team. 5. Those two players go back to their team and two other players come up to the whiteboard. 6. Repeat the process as many times as needed. 7. The team with the most points at the end wins. You choose the prize.	• Wall-mounted whiteboard ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers ***OR*** • Paper • Pens
Chain Writing	1. Give the class a topic to write about. 2. Pass around a piece of lined paper. 3. A student writes one sentence about the topic and then passes it on to the next student. 4. Continue until the topic is completely written about. *Note: Every few students, glance at the paper to see if they have written enough.* 5. Read what the students wrote aloud to the class.	• Lined paper • Pens
Elevator Pitch	1. Have students prepare a 1-2 minute speech summarizing the current topic. 2. Have students give their speeches to the class.	*Optional:* • Paper • Pens
Envelope Sort	1. Provide categories for students and envelopes that have passages/scenarios in them. Example: You could have envelopes for STKYa, STKYp, and STKY success. 2. Students work in partners or groups to sort the passages into the categories. 3. Discuss the correct answers with the class.	• Envelopes with passages or scenarios
Gallery Walk	1. Have students create a poster about a topic. 2. Put the posters up on the walls. 3. Have the class walk around and view each poster.	• Paper • Markers • Tape / Magnets

APPENDIX C—APPLICATION ACTIVITY MENU (cont.)

NAME	DESCRIPTION / STEPS	MATERIALS / PREPARATION
Graffiti Wall	1. Write questions or topics on posters or whiteboards around the room. 2. Have students rotate through and write their response to each question or topic. 3. Review the answers as a class.	• Chart paper • Pens ***OR*** • Wall-mounted whiteboard • Dry-erase markers • Whiteboard erasers
Jigsaw	1. Give each student or pair/group a letter. 2. Have students find a word or phrase from the lesson that starts with that letter. 3. Have students share their words/phrases.	• Paper • Pens • Reading passages *(if necessary)*
Line in the Sand	1. Have the class divide themselves evenly onto opposite sides of the room. 2. Explain that the middle of the room is the "line in the sand" and choosing a side to stand on indicates agreement with that side. *Example: The sides could be disagree/agree, or you could use opposing concepts, such as STKYp and STKY success.* 3. Label each side of the room. 4. Provide prompts to the students. 5. Tell them to stand on the side they think the prompt would fall under. 6. Provide the correct answer.	
My Favorite No	1. Using exit slips, a quiz, or another formative assessment, present your "favorite" wrong answer to the class. 2. Ask the students what is correct or done well first. 3. Ask students what the mistake(s) was. 4. Ask them how to correct it. *Note: You can find the description for exit slips in Appendix D.*	• Completed exit slips ***OR*** • Completed quiz ***OR*** • Another completed formative assessment sample
Pictionary	1. Put key words/phrases/topics that relate to the course and are drawable on slips of paper. 2. Have students take turns picking a slip of paper and drawing the word/phrase/topic on the whiteboard so the rest of the class can see. 3. The rest of the class guesses what the student is drawing.	• Key words/phrases/ topics written on slips of paper • Wall-mounted whiteboard ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers ***OR*** • Key words/phrases/ topics written on slips of paper • Paper • Pens

APPENDIX C—APPLICATION ACTIVITY MENU (cont.)

NAME	DESCRIPTION / STEPS	MATERIALS / PREPARATION
Quick Write	1. Say or put up a prompt. 2. Give students 3-5 minutes to write a quick response. 3. Ask for volunteers to share what they wrote.	• Paper • Pens ***OR*** • Individual whiteboards • Dry erase markers • Whiteboard erasers
Student-Created Questions	1. Have students create 1-3 questions from a selected reading or after a presentation. 2. Students take turns asking and answering each other's questions. 3. Review some of the questions as a class.	• Paper • Pens ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers
Table Discussions	1. Ask/put up a question. 2. Have students work in tables to discuss the question. 3. Have volunteers share their answers.	
T-Charts	1. Have students draw a large T on a piece of paper or their whiteboard. 2. Have students label the two sides of the T at the top. *Example: STKYp and STKY success.* 3. Have students provide examples under each column. 4. Have volunteers share their examples.	• Paper • Pens ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers
Think-Pair-Share	1. Ask/put up a question. 2. Have students think about their answer on their own, silently. 3. Have them share their answers with a partner. 4. Have students volunteer to share their answers with the class.	
Venn Diagram	1. Provide two topics for students to compare and contrast using a Venn diagram. 2. Have students volunteer to share their answers.	• Paper • Pens ***OR*** • Individual whiteboards • Dry-erase markers • Whiteboard erasers
Who/What Am I?	1. Provide students with a key term, person, or topic. 2. Tell them they may not look at it and have to place it on their forehead. 3. Students must then go around asking their classmates questions that will help them guess who/what their slip of paper says. 4. When they think they know they can ask another student. 5. If the student guesses correctly, they can sit down. If they are not correct, they need to keep asking questions until they can guess the correct answer.	• Pre-made key terms, people, or topics on slips of paper
Word/Phrase Sort	1. Give students key terms or phrases written on cards. 2. Students sort the cards into categories. 3. Students share why they sorted the cards the way they did with the class.	• Pre-made key terms or phrases cards

APPENDIX D—FORMATIVE ASSESSMENT MENU

NAME	DESCRIPTION / STEPS	MATERIALS / PREPARATION
ABC Cards	Ask the students a question. Have students hold up a letter card to answer the question. These could be the A/B/C/D cards for multiple choice questions, or a specific letter that correlates to a topic you are working with. *Example: A worker fell from a bucket and wasn't wearing their harness, but didn't get hurt. Hold up an A if you think this would classify as a STKYa, P if you think this would classify as a STKYp, or S if you think this would classify as a STKY success.*	• ABC Cards
Circle the Questions	1. Provide students with a list of questions relating to the topic. 2. Tell them to circle the questions they can't confidently answer. 3. Review the circled questions to inform your instruction and support struggling students.	• Pre-made list of questions • Pens
Exit Slips	1. Pass out a simple, short question printed on a half sheet of paper at the end of class. *Note: The question doesn't have to be multiple choice —it can be open-ended, and it is often better that way.* 2. Give students 3-5 minutes to answer. 3. Collect the exit slips and review them.	• Pre-made questions printed on a half sheet of paper • Pens
Fingers Up	Have students hold up a number of fingers from one to five. One meaning "I need help!" and five meaning "I've got this and could teach others!"	
Hand Thermometer	Have students self-assess their learning by raising their hand to different levels. The closer to the ground the less they understand. The higher in the air the more they understand.	
Quick Nod	These are great for yes or no type questions. Ask a question and students either shake or nod their heads.	
Quiz	Give students a quiz relating to the topics. Review their answers to inform your instruction and support struggling students.	• Instructor-made quizzes ***OR*** • Course quizzes • Pens
Summaries	Have students summarize, in their own words, what they learned during that day/segment. Review these answers to inform your instruction and support struggling students.	• Paper • Pens
Thumbs Up / Thumbs Down	To have students self-assess their understanding, you can ask them to show you thumbs up for if they've got it, thumbs down if they are confused, or even thumbs to the side if they understand a little but could still use some help.	
Traffic Light	Have students use a traffic light format to convey their understanding. Red light for completely lost and no understanding; yellow light for some understanding but needs additional support; green light for complete understanding—no support needed.	• Paper • Red, yellow, and green markers ***OR*** • Individual whiteboards • Red, yellow, and green dry-erase markers • Whiteboard erasers

APPENDIX E—EXTENSION ACTIVITY MENU

These activities are meant to be used for students who are excelling and need extra practice, or if you have extra time in the class and need to fill it in a meaningful way.

NAME	DESCRIPTION / STEPS	MATERIALS / PREPARATION
ABC Books	Have students create an alphabet book with each letter representing something from the concepts taught	• Paper • Markers • Pens
Comic Strips	Have students create a comic strip depicting a concept from the course.	• Paper • Markers • Pens
Create the Test	Have students create a test with a key. It can be multiple choice or have open-ended questions.	• Paper • Pens
One-Pager	Have students create a one-page document summarizing concepts taught. Use unlined paper for this activity. Students should have the following on their page: • Title • Illustration • Quote • Unlined paper • Questions & answers *(minimum of 2)*	• Unlined paper • Markers • Pens
Posters	Have students create posters summarizing the concepts taught.	• Paper • Markers

APPENDIX F—ONLINE LEARNING FOR THE CAPACITY MODEL

The Online Learning for The Capacity Model website provides valuable educational resources. The site is open to everyone. There is no registration requirement. Instructors can use content to supplement their training or share the site with trainees, who can use it as a resource to refresh and extend their learning.

ACCESS THE WEBSITE

Go to: https://learning.thecapacitymodel.com.

ELEARNING COURSES

Courses are available in English, Spanish, and French. Options include:

- Understanding The Capacity Model
- STKY Controls for Crews
- Introduction to Learning Teams

TCM VIDEOS

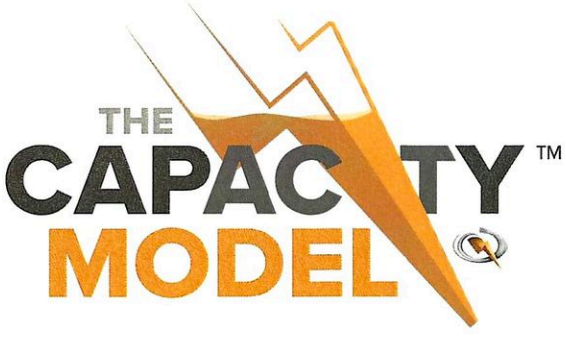
The videos provide viewers an introduction to several elements of The Capacity Model. Some of the options include Quanta Human Performance Principles, STKY, Quanta Energy Wheel, and Quanta's Safety Philosophy.

STKY RE-CREATION VIDEOS

The re-creation videos are based on actual events. Instructors can use these videos to supplement STKY and the Energy Wheel training to reinforce concepts such as using the Energy Wheel, asking capacity questions, identifying preventive and mitigative controls, and classifying STKY events.

RESOURCES

The site's Resources section includes PDFs of printable informational brochures and job aids such as the STKY Discussion Guide.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